Nonparametric Statistics

Lecture notes based on Jung, *Nonparametric Statistics with R* (Ch. 1–12)

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Chapter 1. Introduction to Nonparametric Statistics

1.1 Parametric and nonparametric models

A statistical model is a family $\mathcal{F}$ of distributions assumed to contain the population. It is **parametric** if it is indexed by finitely many real numbers, e.g. the normal family

$$\mathcal{F}_G = \{N(\mu, \sigma^2) : \mu \in \mathbb{R},\ \sigma > 0\}. \tag{1.1}$$

It is **nonparametric** if no finite-dimensional parametrization exists. Typical examples:

$$\mathcal{F}_1 = \{F : \int x^2\, dF < \infty\}, \qquad \mathcal{F}_{\text{all}} = \{\text{all } F\},$$

$$\mathcal{F}_2 = \{F : F \text{ has a density symmetric about some } \theta\}.$$

Both $\mathcal{F}_G \subset \mathcal{F}_1$ and $\mathcal{F}_G \subset \mathcal{F}_2$. A model with a finite-dimensional part of interest and an infinite-dimensional nuisance part, such as $y_i = \beta_0 + \beta_1 x_i + \epsilon_i$ with $\epsilon_i \overset{\text{i.i.d.}}{\sim} F$ unrestricted, is called **semi-parametric**.

In a nonparametric model a **parameter** is a functional $\theta = \theta(F)$: the mean $\int x\, dF$, the median, a quantile, the density f, or a regression curve $m(x) = E(Y \mid X = x)$. A procedure whose validity (e.g. type I error) holds for every F in a very large family is **distribution-free**.

Three tools:

1. **Ranks, signs and permutation tests** (Part I): conditional inference given the observed values, exact under exchangeability.
2. **The bootstrap** (Part II): replace the unknown F by an estimate and compute sampling distributions by simulation.
3. **Smoothing** (Part II): estimate a function $f(x)$ or $m(x)$ by averaging the observations near x.

1.2 Review of parametric inference

Motivation. We recall the t-test and likelihood framework to see which assumptions each rank method drops.

Definition (Quantile)

For a distribution function F and $p \in (0,1)$, the p-quantile is $Q(p) = \min\{x : F(x) \geq p\}$. For the standard normal, $z_\alpha := \Phi^{-1}(1-\alpha)$.

Proposition 1.1

Let F be the distribution function of an integer-valued X (e.g. $X \sim B(n,p)$). Then $q = Q(p)$ is the smallest integer with $F(q) \geq p$, and $F(q-1) < p \leq F(q)$. In particular $P(X \leq q) \geq p$ and $P(X \geq q) > 1 - p$.

Proof. F is nondecreasing, right-continuous and constant on each $[k, k+1)$, so $\{x : F(x) \geq p\} = [q, \infty)$ with q the smallest integer satisfying $F(q) \geq p$. Minimality gives $F(q-1) < p$. Then $P(X \geq q) = 1 - F(q-1) > 1 - p$. ∎

If $X \sim N(\mu, \sigma^2)$, $F_X(x) = \Phi((x-\mu)/\sigma)$. If $X \sim B(n,p)$, $EX = np$, $\operatorname{Var} X = np(1-p)$, and for $np \geq 10$ roughly $X \approx N(np, np(1-p))$ by the CLT.

Estimation. An estimator $\hat{\theta}$ is judged by its bias $E\hat{\theta} - \theta$, standard error $\mathrm{se}(\hat{\theta}) = \sqrt{\mathrm{Var}\,\hat{\theta}}$ and mean squared error

$$\mathrm{MSE}(\hat{\theta}) = E(\hat{\theta} - \theta)^2 = \mathrm{bias}(\hat{\theta})^2 + \mathrm{Var}(\hat{\theta}).$$

Under $\mathcal{F}_G$, $\bar{X}_n$ is the best unbiased estimator of μ, and $T = \sqrt{n}(\bar{X}_n - \mu)/s \sim t_{n-1}$ gives the exact interval $\bar{X}_n \pm t_{\alpha/2,n-1}s/\sqrt{n}$; outside $\mathcal{F}_G$ both properties are lost.

Testing. A test rejects when a statistic T falls in a region with $P_{H_0}(\text{reject}) \le \alpha$ (**level**); we want large **power** $P_{H_1}(\text{reject})$. The p**-value** is the H_0-probability of a value of T at least as extreme as observed; $p \le \alpha$ iff T lies in the level-α region. Nonparametric problems need a statistic whose **null distribution** does not depend on F.

Proposition 1.2 (Duality of tests and intervals)

Suppose for each θ_0 there is a level-α test ϕ_{θ_0} of $H_0 : \theta = \theta_0$. Then $C = \{\theta_0 : \phi_{\theta_0} \text{ does not reject}\}$ is a $1-\alpha$ confidence set. Conversely, if C is a $1-\alpha$ confidence set, rejecting $H_0 : \theta = \theta_0$ when $\theta_0 \notin C$ is a level-α test.

Proof. If θ is the true value, $P(\theta \in C) = P(\phi_\theta \text{ does not reject}) \ge 1-\alpha$. Conversely $P_{\theta_0}(\theta_0 \notin C) \le \alpha$. ■

The central limit theorem.

Theorem 1.3 (CLT)

If $X_1, \dots, X_n \overset{\text{i.i.d.}}{\sim} F$ with mean μ and variance $0 < \sigma^2 < \infty$, then $\sqrt{n}(\bar{X}_n - \mu)/\sigma \xrightarrow{d} N(0,1)$.

Proof [Supplement]. Let $Y_i = (X_i - \mu)/\sigma$ and $\varphi(t) = Ee^{itY_1}$. Since $EY_1^2 < \infty$, $\varphi(t) = 1 - \frac{t^2}{2} + o(t^2)$. By independence $Ee^{itZ_n} = \varphi(t/\sqrt{n})^n = (1 - \frac{t^2}{2n} + o(\frac{1}{n}))^n \to e^{-t^2/2}$, the characteristic function of $N(0,1)$; Lévy's continuity theorem concludes. ■

So t-procedures need only the *statistic* to be approximately normal. But the CLT gives no rate: convergence is slow for skewed F and fails when $\sigma^2 = \infty$ (Cauchy, slash), the first motivation for nonparametric methods.

1.3 Why nonparametric methods

- *Validity under weak assumptions.* Rank, sign and permutation tests control the type I error exactly for every F in a large family, at any sample size.
- *Computation instead of derivation.* Permutation and bootstrap distributions replace formulas that are unknown or valid only asymptotically.
- *Flexibility.* Function-valued targets (f, m) are estimated without assuming a shape.
- *Robustness.* Procedures based on ranks, signs or medians are insensitive to outliers.

Example 1.1 (Effect of one outlier)

Take 2.1, 2.4, 2.6, 2.9, 3.0, 3.2, 3.5, 25. With/without the last value: mean 5.59 / 2.81, standard deviation 7.86 / 0.48, median 2.95 / 2.9, and **median absolute deviation** MAD $= \mathrm{median}_i|x_i - \mathrm{median}_j\, x_j|$ 0.45 / 0.30. One point dominates the mean and SD; the median and MAD barely move.

Distribution-free methods keep their properties under heavy tails (where outliers come from), so they are typically robust; the converse need not hold.

1.4 Selected exercises

Motivation. Exercises on quantiles and CLT-based inference, and on exact pivots that replace the normal approximation.

Exercise 1.1 (Normal quantiles)

Express the quantile function $Q(p)$ of $N(\mu, \sigma^2)$ through Φ, and find the median $Q(0.5)$ and the quartiles $Q(0.25)$, $Q(0.75)$.

Solution.

Step 1. $F(x) = \Phi((x-\mu)/\sigma)$ is continuous and strictly increasing on $\mathbb{R}$ with limits 0 and 1. So for $p \in (0,1)$ there is a unique x_p with $F(x_p) = p$, and $\{x : F(x) \geq p\} = [x_p, \infty)$. Hence $Q(p) = x_p$.

Step 2. $F(x_p) = p$ means $(x_p - \mu)/\sigma = \Phi^{-1}(p)$, so

$$Q(p) = \mu + \sigma\, \Phi^{-1}(p).$$

Step 3. Since ϕ is even, $\Phi(-z) = 1 - \Phi(z)$, so $\Phi^{-1}(1-p) = -\Phi^{-1}(p)$ and $\Phi^{-1}(0.5) = 0$. With $\Phi^{-1}(0.75) = 0.6745$,

$$Q(0.5) = \mu, \qquad Q(0.25) = \mu - 0.6745\,\sigma, \qquad Q(0.75) = \mu + 0.6745\,\sigma.$$

So the normal IQR is $1.349\,\sigma$: the source of the divisor 1.34 in Section 11.2 and of the MAD factor $1/0.6745 = 1.4826$ (Section 10.3.1). ■

Exercise 1.5 (Exponential waiting times)

Let $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} \text{Exp}(\theta)$ with mean $\theta > 0$, i.e. density $\theta^{-1} e^{-x/\theta}$ for $x > 0$.
(a) Use the CLT to find the approximate distribution of $\bar{X}_n$.
(b) The waiting time for a bus is modeled this way, and $n = 40$ students waited $\bar{x} = 7$ minutes on average. Give an approximate 95% confidence interval for θ, and test $H_0 : \theta = 10$ against $H_1 : \theta < 10$; report the p-value and the conclusion at $\alpha = 0.01$.

Solution.

(a) With $y = x/\theta$, $EX = \theta \int_0^\infty y e^{-y}\, dy = \theta$ and $EX^2 = \theta^2 \int_0^\infty y^2 e^{-y}\, dy = 2\theta^2$, so $\text{Var}\, X = \theta^2$. By Theorem 1.3, $\sqrt{n}(\bar{X}_n - \theta)/\theta \xrightarrow{d} N(0,1)$, i.e. $\bar{X}_n$ is approximately $N(\theta, \theta^2/n)$.

(b) *Interval.* By the LLN $\bar{X}_n/\theta \xrightarrow{p} 1$, so by Slutsky $\sqrt{n}(\bar{X}_n - \theta)/\bar{X}_n \xrightarrow{d} N(0,1)$, giving

$$\bar{x} \pm 1.96\, \frac{\bar{x}}{\sqrt{n}} = 7 \pm 1.96 \cdot \frac{7}{\sqrt{40}} = 7 \pm 2.17 = (4.83,\ 9.17).$$

Test. Under H_0 the SD $\theta_0/\sqrt{n}$ is known, so use $Z = \sqrt{n}(\bar{X}_n - \theta_0)/\theta_0$, rejecting for small Z: $z = \sqrt{40}(7-10)/10 = -1.90$, $p = \Phi(-1.90) = 0.029 > 0.01$. Not rejected at 1% (rejected at 5%). ■

Exercise 1.7 (Exact tests under a model)

Can the test of Exercise 1.5 be carried out exactly, without the CLT, if the population is assumed to be (a) Exp(θ), or (b) Uniform$(0, 2\theta)$ (which also has mean θ)?

Solution.

(a)

Step 1. (A pivot.) $Y_i = 2X_i/\theta$ has the χ^2_2 density $\frac{1}{2}e^{-y/2}$, with m.g.f. $(1-2t)^{-1}$, $t < \frac{1}{2}$; so $\sum_i Y_i$ has m.g.f. $(1-2t)^{-n}$, that of χ^2_{2n}:

$$V(\theta) = \frac{2n\bar{X}_n}{\theta} \sim \chi^2_{2n} \qquad \text{for every } \theta > 0.$$

Step 2. (Test.) Under H_0, $V(\theta_0) \sim \chi^2_{2n}$; for $\theta < \theta_0$, $V(\theta_0) = (\theta/\theta_0)V(\theta)$ is stochastically smaller, so reject for small $V(\theta_0)$: $p = P(\chi^2_{2n} \le 2n\bar{x}/\theta_0)$. Here $v = 56$ and $p = P(\chi^2_{80} \le 56) = 0.019$, not significant at 1%.

Step 3. (Interval.) With q_p the p-quantile of χ^2_{2n}, $P(q_{0.025} \le V(\theta) \le q_{0.975}) = 0.95$ gives the exact interval $\left(2n\bar{X}_n/q_{0.975},\ 2n\bar{X}_n/q_{0.025}\right) = (5.25,\ 9.80)$. It lies right of the CLT interval and the p-value is smaller, reflecting the right skew of $\bar{X}_n$.

(b)

Step 1. (A pivot.) Let $M = \max_i X_i$. For $0 \le m \le 2\theta$, $P(M \le m) = (m/2\theta)^n$, so $U(\theta) = (M/2\theta)^n$ has $P(U(\theta) \le u) = P(M \le 2\theta u^{1/n}) = u$: $U(\theta) \sim$ Uniform$(0, 1)$.

Step 2. (Test.) Under $\theta < \theta_0$ all observations lie below 2θ, so small M supports H_1: $p = P_{\theta_0}(M \le m_{\text{obs}}) = \min\{1, (m_{\text{obs}}/2\theta_0)^n\}$. Under H_0, $p = U(\theta_0)$ is uniform, so $P(p \le \alpha) = \alpha$: exact.

Both exact tests use a model-specific statistic ($\bar{X}_n$ with the χ^2 law, or M; $\bar{x}$ alone does not suffice for (b)) and are exact only under their own model, a dependence nonparametric methods avoid. ■

Chapter 2. Distribution-Free Inference for One Population

Setting

$X_1, \dots, X_n \overset{\text{i.i.d.}}{\sim} F$ with median θ; test $H_0 : \theta = \theta_0$, estimate θ and build intervals, assuming F continuous (sign test) or continuous and symmetric about θ (signed-rank test).

2.1 The sign test

Motivation. For any continuous F the number of observations above the median is exactly binomial.

Definition (Sign statistic)

$S = S(\theta_0) = \sum_{i=1}^{n} I(X_i > \theta_0)$, the number of observations above θ_0.

Proposition 2.1

If $P(X_i = \theta_0) = 0$ and θ_0 is the median of F, then $S \sim B(n, \frac{1}{2})$ for every such F.

Proof. The indicators $I(X_i > \theta_0)$ are i.i.d. Bernoulli with success probability $P(X_1 > \theta_0) = 1 - F(\theta_0) = \frac{1}{2}$. ■

So the sign test is distribution-free. Under $\theta > \theta_0$, $p_1 = P(X > \theta_0) > \frac{1}{2}$: reject for large S, $p = P(S \geq s_{\text{obs}})$; two-sided $p = 2\min\{P(S \geq s_{\text{obs}}), P(S \leq s_{\text{obs}})\}$.

Proposition 2.2 (Rejection region)

The smallest c with $P(S \geq c) \leq \alpha$ under H_0 is $c = Q_B(1-\alpha) + 1$, where Q_B is the quantile function of $B(n, \frac{1}{2})$.

Proof. $P(S \geq c) \leq \alpha \iff F_B(c-1) \geq 1-\alpha$, and the smallest integer $c-1$ with this property is $Q_B(1-\alpha)$ (Proposition 1.1). ■

The attained size is usually below α (discreteness). For large n, $(S - n/2)/\sqrt{n/4} \dot{\sim} N(0,1)$.

Ties. Discard observations equal to θ_0 and reduce n; given the number discarded, the remaining signs are i.i.d. fair coins when $P(X < \theta_0) = P(X > \theta_0)$.

Example 2.1

Twelve paired differences (after − before) have 9 positive signs. For $H_1 : \theta > 0$, $p = P(S \geq 9) = (220 + 66 + 12 + 1)/4096 = 0.073$.

Power. Under an alternative $S \sim B(n, p_1)$, $p_1 = P(X > \theta_0)$, so the power is $1 - F_B(c-1; n, p_1)$; it depends on the shape of F.

Example 2.2 (Sign test vs. z-test)

Let $n = 20$, $\theta_0 = 0$, $\theta = 0.5$, standard deviation 1. The size-≤ 0.05 sign test rejects when $S \geq 15$ (actual size 0.021).

- Normal $N(0.5, 1)$: $p_1 = \Phi(0.5) = 0.69$, power 0.38; the z-test has power $\Phi(0.5\sqrt{20} - 1.645) = 0.72$.
- Laplace, same mean and variance (scale $1/\sqrt{2}$): $p_1 = 1 - \frac{1}{2}e^{-0.5\sqrt{2}} = 0.75$, power 0.63; z-test still about 0.72.

The sign test gains with heavy tails (Section 2.3).

2.1.1 Confidence interval for the median

Proposition 2.3

Let F be continuous, $X_{(1)} < \cdots < X_{(n)}$ the order statistics and $l = Q_B(\alpha/2)$, $u = n - l + 1$. Then

$$P(X_{(l)} \leq \theta < X_{(u)}) = \sum_{k=l}^{n-l} \binom{n}{k} 2^{-n} \geq 1 - \alpha.$$

Proof. $X_{(l)} \leq \theta$ iff at most $n-l$ observations exceed θ, and $\theta < X_{(u)}$ iff at least $n-u+1 = l$ do. So the event is $\{l \leq S(\theta) \leq n-l\}$ with $S(\theta) \sim B(n, \frac{1}{2})$. By symmetry $P(S \leq l-1) = P(S \geq n-l+1)$, and each is $\leq \alpha/2$ by the choice of l (if $P(S \leq Q_B(\alpha/2)) = \alpha/2$ exactly, take $l = Q_B(\alpha/2) + 1$). ■

For $n = 15$, $\alpha = 0.05$: $l = 4$, and $(X_{(4)}, X_{(12)})$ has coverage 0.965. For large n, $S \dot{\sim} N(n/2, n/4)$ gives

$$l \approx \frac{n}{2} - z_{\alpha/2}\frac{\sqrt{n}}{2}, \qquad u \approx \frac{n}{2} + z_{\alpha/2}\frac{\sqrt{n}}{2} + 1 \qquad \text{(rounded to integers)},$$

an asymptotically valid distribution-free interval. As the level decreases it shrinks to the sample median.

For any quantile $Q(p)$: $\#\{X_i \leq Q(p)\} \sim B(n, p)$, so $(X_{(l)}, X_{(u)})$ covers $Q(p)$ with probability $\sum_{k=l}^{u-1} \binom{n}{k} p^k (1-p)^{n-k}$.

2.2 The Wilcoxon signed-rank test

Motivation. Under symmetry the ranks of $|X_i|$ add information while keeping the test distribution-free.

Assume now that F is continuous and symmetric about θ. Let $D_i = X_i - \theta_0$, $V_i = I(D_i > 0)$, and let R_i be the rank of $|D_i|$ among $|D_1|, \ldots, |D_n|$.

Definition (Signed-rank statistic)

$SR_+ = \sum_{i=1}^{n} V_i R_i$, the sum of the ranks of the positive deviations.

Lemma 2.4

Under H_0 (symmetry about θ_0), $(V_1, \ldots, V_n)$ are i.i.d. Bernoulli($\frac{1}{2}$) and independent of

$(|D_1|, \dots, |D_n|)$. Consequently

$$SR_+ \overset{d}{=} \sum_{k=1}^{n} kB_k, \qquad B_k \overset{\text{i.i.d.}}{\sim} \text{Bernoulli}(\tfrac{1}{2}).$$

Proof. For a symmetric continuous D, $\text{sgn}(D)$ and $|D|$ are independent with $P(D > 0) = \frac{1}{2}$: $P(D > 0, |D| \le t) = P(0 < D \le t) = \frac{1}{2}P(|D| \le t)$. Independence across i gives the first claim. Given the ranks, (R_i) is a permutation π of $(1, \dots, n)$ and $\sum_i V_i R_i = \sum_k k V_{\pi^{-1}(k)}$, with $(V_{\pi^{-1}(k)})_k$ again i.i.d. fair coins. ■

Proposition 2.5

Under H_0, the null distribution of SR_+ depends only on n, is symmetric about $n(n+1)/4$, and

$$E(SR_+) = \frac{n(n+1)}{4}, \qquad \text{Var}(SR_+) = \frac{n(n+1)(2n+1)}{24}, \qquad \frac{SR_+ - E(SR_+)}{\sqrt{\text{Var}(SR_+)}} \xrightarrow{d} N(0,1).$$

Proof [Supplement]. From Lemma 2.4, $E = \frac{1}{2}\sum k$ and $\text{Var} = \frac{1}{4}\sum k^2$. Replacing B_k by $1 - B_k$ (same law) maps SR_+ to $\frac{n(n+1)}{2} - SR_+$, giving symmetry. For normality use Lyapunov's CLT with $U_k = kB_k$: $\sum_k E|U_k - k/2|^3 = \sum k^3/8 = O(n^4)$, while $s_n^3 = \text{Var}(SR_+)^{3/2} \asymp n^{9/2}$, so the Lyapunov ratio is $O(n^{-1/2}) \to 0$. ■

The exact null distribution enumerates the 2^n equally likely sign patterns (*sign-flip* principle): $p = P(SR_+ \ge sr_{\text{obs}})$ against $\theta > \theta_0$, $P(SR_+ \le sr_{\text{obs}})$ against $\theta < \theta_0$, and twice the smaller one two-sided (the null law is symmetric).

Example 2.3

Deviations -0.3, 0.8, 1.5, 2.2, 2.9: absolute ranks $1, \dots, 5$, only rank 1 negative, so $SR_+ = 14$. Of the 32 sign patterns, $SR_+ \ge 14$ for "all positive" (15) and "only rank 1 negative" (14): $p = 2/32 = 0.0625$. The sign test gives $P(S \ge 4) = 6/32 = 0.19$.

Ties. Drop zero deviations; give tied $|D_i|$ mid-ranks. The sign-flip distribution of the actual mid-ranks is still exact conditionally; for the normal approximation use $\text{Var}(SR_+) = \frac{1}{4}\sum_i R_i^2$.

2.2.1 Walsh averages, the Hodges–Lehmann estimator and intervals

The $N = n(n+1)/2$ **Walsh averages** are $W_{ij} = (X_i + X_j)/2$, $1 \le i \le j \le n$.

Proposition 2.6

If no $|X_k - \theta|$ are tied and no W_{ij} equals θ, then $SR_+(\theta) = \#\{(i \le j) : W_{ij} > \theta\}$.

Proof [Supplement]. Let $d_k = X_k - \theta$, so $W_{ij} > \theta \iff d_i + d_j > 0$. Assign each pair $\{i, j\}$ to the $k \in \{i, j\}$ with larger $|d_k|$; then $\text{sgn}(d_i + d_j) = \text{sgn}(d_k)$, and k receives the pairs $\{k, m\}$ with $|d_m| \le |d_k|$, i.e. R_k pairs. Hence $\#\{W_{ij} > \theta\} = \sum_{k:d_k>0} R_k$. ■

So SR_+ is the sign statistic of the Walsh averages, and Proposition 2.3 transfers.

Corollary 2.7

Let $W_{(1)} < \cdots < W_{(N)}$ and $l = Q_{SR}(\alpha/2)$ (quantile of the null distribution of SR_+), $u = N - l + 1$. Then $P(W_{(l)} \le \theta < W_{(u)}) \ge 1 - \alpha$. The point estimator obtained as the level shrinks is the **Hodges–Lehmann estimator** $\hat{\theta}_{HL} = \text{median}_{i \le j} W_{ij}$.

Proof. $\theta < W_{(a)} \iff SR_+(\theta) > N - a$ and $\theta \ge W_{(a)} \iff SR_+(\theta) \le N - a$; at the true θ, $SR_+(\theta)$ has the null law, symmetric about $N/2$. Repeat the proof of Proposition 2.3. ■

Paired data: apply these methods to the differences.

2.3 Comparing the t-, sign and signed-rank procedures

Validity. The sign test is exact for continuous F, the signed-rank test for continuous symmetric F, the t-test only under normality. For asymmetric F the t-test does not test the median. Nominal 5% one-sided t-test:

- light tails (uniform): size slightly above 5% for $n \approx 5$, essentially correct by $n \approx 10$;
- infinite variance (slash, Cauchy): size well below 5%, not improving with n (CLT fails);
- skewed (exponential, testing the mean): slow convergence, size noticeably above 5% even at $n = 100$;
- skewed, testing the median with t: centered at the wrong value, meaningless.

Only valid tests (or ones with corrected critical values) are compared on power, which depends on F, n and α.

Efficiency. $RE(\hat{\theta}_1, \hat{\theta}_2) = \text{MSE}(\hat{\theta}_2)/\text{MSE}(\hat{\theta}_1)$; its limit is the **asymptotic relative efficiency** (ARE). For tests the (Pitman) ARE is the limiting ratio N_2/N_1 of sample sizes for equal power at $\theta_n \to \theta_0$; here it equals the ARE of $\bar{X}_n$, median X_i, $\hat{\theta}_{HL}$.

For intervals of equal coverage, efficiency is $1/\text{length}^2$. If $\hat{\theta} \dot{\sim} N(\theta, v/n)$, $\hat{\theta} \pm z_{\alpha/2}\sqrt{v/n}$ has length

$$2z_{\alpha/2}\sqrt{v/n} \approx 2z_{\alpha/2}\sqrt{\text{MSE}(\hat{\theta})}.$$

so interval ARE is the ratio of MSEs: estimators, tests and intervals rank alike. E.g. for a normal population $ARE(\bar{X}_n, \text{median } X_i) = \pi/2 \approx 1.57$.

Proposition 2.8

Let F have density f symmetric about θ, with $f(\theta) > 0$ and variance σ^2 (possibly infinite). Then

$$\sqrt{n}(\bar{X}_n - \theta) \xrightarrow{d} N(0, \sigma^2), \qquad \sqrt{n}(\text{median } X_i - \theta) \xrightarrow{d} N\Big(0, \frac{1}{4f(\theta)^2}\Big),$$

$$\sqrt{n}(\hat{\theta}_{HL} - \theta) \xrightarrow{d} N\Big(0, \frac{1}{12(\int f^2)^2}\Big).$$

Hence

$$ARE(\text{sign}, t) = 4\sigma^2 f(\theta)^2, \qquad ARE(SR, t) = 12\sigma^2\Big(\int f^2\Big)^2, \qquad ARE(\text{sign}, SR) = \frac{f(\theta)^2}{3(\int f^2)^2}.$$

Proof for the median; sketch for HL. $P(\text{median } X_i \le \theta + t/\sqrt{n}) = P(\#\{X_i \le \theta + t/\sqrt{n}\} \ge \lceil n/2 \rceil)$, the count being $B(n, p_n)$, $p_n = \frac{1}{2} + f(\theta)t/\sqrt{n} + o(n^{-1/2})$; the normal approximation gives $\Phi(2f(\theta)t)$, variance $1/(4f(\theta)^2)$. For $\hat{\theta}_{HL}$, by Proposition 2.6, $\hat{\theta}_{HL} \le \theta + t/\sqrt{n}$ iff $SR_+(\theta + t/\sqrt{n}) \le N/2$; this statistic has mean $N/2 - n^{3/2}t\int f^2 + o(n^{3/2})$ and variance $\sim n^3/12$, giving $\Phi(t\sqrt{12}\int f^2)$. ■

F	ARE(sign, t)	ARE(SR, t)	ARE(sign, SR)
Uniform	$1/3$	1	$1/3$
Normal	$2/\pi \approx 0.64$	$3/\pi \approx 0.955$	$2/3$
Logistic	$\pi^2/12 \approx 0.82$	$\pi^2/9 \approx 1.10$	$3/4$
Laplace	2	1.5	$4/3$
Cauchy, slash	∞	∞	finite

(Check: $N(0,1)$ has $f(0)^2 = \frac{1}{2\pi}$, $\int \phi^2 = \frac{1}{2\sqrt{\pi}}$; Laplace $\frac{1}{2}e^{-|x|}$ has $\sigma^2 = 2$, $f(0) = \frac{1}{2}$, $\int f^2 = \frac{1}{4}$.) Over all symmetric F, $ARE(SR, t) \ge 0.864$ (Hodges–Lehmann): signed-rank costs little under normality and can gain a lot; the sign test wins only for very heavy tails.

2.4 Selected exercises

Motivation. A sign test for a non-median quantile, and the tie-corrected signed-rank moments used by software.

Exercise 2.4 (A sign test for the 10th percentile)

Ten foldable phones were folded until failure. The numbers of folds (in units of 10^4) were

$$7,\ 9,\ 12,\ 39,\ 28,\ 11,\ 22,\ 17,\ 14,\ 9.$$

Can one claim that at least 90% of such phones survive 10^5 folds? Use a sign-type test at level 0.1, and give a 90% confidence bound for the 10th percentile of the number of folds.

Solution.

Step 1. (Hypotheses.) Let $\xi = Q(0.1)$ of the continuous F. "At least 90% survive 10 units" means $P(X < 10) \le 0.1$, i.e. $\xi \ge 10$; as the claim, it is the alternative: $H_0 : P(X < 10) = 0.1$ $(\xi = 10)$ vs. $H_1 : P(X < 10) < 0.1$ $(\xi > 10)$.

Step 2. (Test.) $K = \#\{i : X_i < 10\} \sim B(10, 0.1)$ under H_0 (Proposition 2.1 with $\frac{1}{2} \to 0.1$); small K supports H_1. Here $7, 9, 9$ give $k = 3$ and

$$p = P(K \le 3) = 0.987.$$

H_0 is not rejected: three early failures exceed the one expected under H_0.

Step 3. (Confidence bound.) By Section 2.1.1, $J = \#\{i : X_i \le \xi\} \sim B(10, 0.1)$, and (no ties a.s.) $\xi < X_{(u)} \iff J \le u - 1$. Since $P(J \le 1) = 0.736 < 0.9 \le 0.930 = P(J \le 2)$, take $u = 3$:

$$P(\xi < X_{(3)}) = 0.930 \ge 0.9.$$

$X_{(3)} = 9$ (the rounding tie $9, 9$ is harmless). A two-sided interval is impossible: $P(\xi < X_{(1)}) = 0.9^{10} = 0.349 > 0.05$.

With confidence $\ge 90\%$ the 10th percentile is below 9×10^4 folds: the claim is not supported. ■

Exercise 2.8 (Signed-rank moments with zeros and ties)

To test $H_0 : \theta = \theta_0$, let $D_i = X_i - \theta_0$. Rank all n values $|D_i|$ (zeros included, occupying the lowest ranks) with mid-ranks for ties; SR_+ is the rank sum of the positive D_i. Suppose t_0 of the D_i are zero and the nonzero $|D_i|$ form tie groups of sizes $t_1, \dots, t_g$ (e.g. $D = 0, 0, 1, 1, 2, 3, 3, 3$: $n = 8$, $t_0 = 2$, sizes $2, 1, 3$). Show that under H_0 (symmetry of D about 0), conditionally on $|D_1|, \dots, |D_n|$,

$$E(SR_+) = \frac{n(n+1) - t_0(t_0+1)}{4},$$
$$\mathrm{Var}(SR_+) = \frac{n(n+1)(2n+1) - t_0(t_0+1)(2t_0+1)}{24} - \frac{\sum_l (t_l^3 - t_l)}{48}.$$

Solution.

Step 1. (Signs given magnitudes.) Symmetry gives, for $A \subset (0, \infty)$, $P(D > 0, |D| \in A) = P(D < 0, |D| \in A)$, and the two events partition $\{|D| \in A\}$; so $P(D > 0 \mid |D|) = \frac{1}{2}$ on $\{|D| > 0\}$. Given $(|D_j|)$, the $V_i = I(D_i > 0)$, $i \in \mathcal{N} = \{i : D_i \neq 0\}$, are i.i.d. Bernoulli($\frac{1}{2}$), $V_i = 0$ otherwise, and the ranks are fixed: $SR_+ = \sum_{i \in \mathcal{N}} V_i R_i$.

Step 2. (Mean.) $E(SR_+ \mid |D|) = \frac{1}{2} \sum_{i \in \mathcal{N}} R_i$. Zeros take positions $1, \dots, t_0$, and mid-ranks preserve sums over tie groups, so

$$\sum_{i \in \mathcal{N}} R_i = \sum_{k=t_0+1}^{n} k = \frac{n(n+1) - t_0(t_0+1)}{2}.$$

Step 3. (A tie group.) A group in positions $a+1, \dots, a+t$ gets $\bar{r} = a + \frac{t+1}{2}$, and

$$\sum_{k=a+1}^{a+t} k^2 - t\bar{r}^2 = \sum_{k=a+1}^{a+t} (k - \bar{r})^2 = \sum_{j=1}^{t} \left(j - \frac{t+1}{2}\right)^2 = \frac{t(t^2-1)}{12}.$$

Step 4. (Variance.) $\mathrm{Var}(SR_+ \mid |D|) = \frac{1}{4} \sum_{i \in \mathcal{N}} R_i^2$, and by Step 3

$$\sum_{i \in \mathcal{N}} R_i^2 = \sum_{k=t_0+1}^{n} k^2 - \sum_l \frac{t_l^3 - t_l}{12} = \frac{n(n+1)(2n+1) - t_0(t_0+1)(2t_0+1)}{6} - \sum_l \frac{t_l^3 - t_l}{12}.$$

Divide by 4. With $t_0 = 0$ and no ties this is Proposition 2.5.

Check: nonzero mid-ranks $3.5, 3.5, 5, 7, 7, 7$ have sum 33 and sum of squares 196.5, so $E = 16.5 = (72 - 6)/4$, $\mathrm{Var} = 49.125 = (1224 - 30)/24 - 30/48$. ■

Chapter 3. Permutation Tests

3.1 Two sources of randomness

- **Randomization model.** N fixed units, m randomly assigned to treatment 1 and $n = N - m$ to treatment 2; responses are fixed, only the assignment is random.
- **Population model.** $X_1, \dots, X_m \overset{\text{i.i.d.}}{\sim} F$ and $Y_1, \dots, Y_n \overset{\text{i.i.d.}}{\sim} G$ are random samples. Inference concerns F and G, e.g. $H_0 : F = G$.

Both use the *permutation* (*randomization*) distribution of a statistic over all re-assignments of the observed values.

3.2 The randomization test

Under H_0 (no treatment effect) each response is the same under either treatment, and the observed assignment is one of $K = \binom{N}{m}$ equally likely ones, all yielding the same pooled responses. With $d_1, \dots, d_K$ the values of a statistic D (e.g. $\bar{Y} - \bar{X}$) over all assignments,

$$p = \frac{1}{K} \sum_{k=1}^{K} I(d_k \geq D_{\text{obs}}). \tag{3.1}$$

Two-sided: use $|d_k| \geq |D_{\text{obs}}|$, not twice the smaller one-sided value (the distribution need not be symmetric).

Example 3.1

Five patients; two random ones get a new treatment (recovery times $3, 5$), the others have $8, 9, 12$. With $D = \bar{Y}_{\text{old}} - \bar{X}_{\text{new}}$ and pooled total 37, a treated pair with sum s gives $D = (37 - s)/3 - s/2$, decreasing in s. The 10 pair sums are $8, 11, 12, 13, 14, 15, 17, 17, 20, 21$; the observed 8 is smallest, so $D_{\text{obs}} = 5.67$ is largest and $p = 1/10$.

Proposition 3.1 (Exactness)

Under H_0, if the d_k have no ties and $\alpha = j/K$, the randomization test rejecting when $p \leq \alpha$ has type I error exactly α. With ties it is at most α.

Proof. Under H_0 the multiset $\{d_k\}$ does not depend on the realized assignment, which is uniform, so D_{obs} is a uniform element of it. Without ties, $p \leq j/K$ iff D_{obs} is among the j largest, probability j/K; ties only increase p. ■

Proposition 3.2 (Equivalent statistics)

Over all re-assignments of a fixed pooled sample, $\sum_j Y_j$, $\bar{Y} - \bar{X}$ and the pooled two-sample t-statistic are strictly increasing functions of each other, hence give identical permutation p-values.

Proof. The pooled total $\sum X_i + \sum Y_j$ is fixed, so $\bar{Y} - \bar{X} = \frac{N}{mn} \sum Y_j - \frac{\text{total}}{m}$ is increasing in $\sum Y_j$. The total sum of squares SST is also fixed, and $SST = (N-2)s_p^2 + \frac{mn}{N} D^2$ with $D = \bar{Y} - \bar{X}$. Hence

$$T = \frac{D}{\sqrt{\frac{1}{m} + \frac{1}{n}} \sqrt{(SST - \frac{mn}{N} D^2)/(N-2)}} = g(D), \qquad g'(D) \propto \frac{SST}{(SST - \frac{mn}{N} D^2)^{3/2}} > 0.$$

■

A difference of medians or trimmed means is *not* monotone in D and can give different (for skewed or heavy-tailed data often more powerful) tests.

3.3 Permutation tests in the population model

Motivation. Under $H_0 : F = G$ the pooled sample is exchangeable, so permutation tests remain valid for random samples.

Definition (Exchangeability)

$(Z_1, \dots, Z_N)$ is **exchangeable** if $(Z_{\pi(1)}, \dots, Z_{\pi(N)}) \stackrel{d}{=} (Z_1, \dots, Z_N)$ for every permutation π.

I.i.d. variables are exchangeable; so are equicorrelated normals $N_N(\mu\mathbf{1}, \sigma^2[(1-\rho)I + \rho\mathbf{1}\mathbf{1}'])$. Independent samples from $N(\mu, 1)$ and $N(\mu, 4)$ are *not*.

Theorem 3.3 (Validity of permutation tests)

Let $Z = (Z_1, \dots, Z_N)$ be exchangeable under H_0, let $\mathcal{G}$ be a group of permutations, and let $p(Z) = \frac{1}{|\mathcal{G}|} \sum_{g \in \mathcal{G}} I(T(gZ) \geq T(Z))$. Then $P_{H_0}(p(Z) \leq \alpha) \leq \alpha$ for every α, with equality when $\alpha|\mathcal{G}|$ is an integer and there are no ties. The statement holds conditionally on the multiset of observed values, hence also unconditionally.

Proof. Let U be uniform on $\mathcal{G}$, independent of Z; $UZ \stackrel{d}{=} Z$. Since $\mathcal{G}$ is a group, $\{gUZ : g \in \mathcal{G}\} = \{gZ\}$, so $T(UZ)$ is a uniform member of the fixed multiset $\{T(gZ)\}$ and lies among its top j (ties counted generously) with probability $\leq j/|\mathcal{G}|$. Hence $P(p(Z) \leq \alpha) = P(p(UZ) \leq \alpha) \leq \alpha$; average over the conditioning. ■

Two samples under $H_0 : F = G$: the pooled observations are i.i.d., and the re-assignments are the $\binom{N}{m}$ splits (each from $m!\,n!$ permutations). This is **conditional inference**: values fixed, labels random.

3.4 Random (Monte Carlo) permutation tests

When $\binom{N}{m}$ is too large, draw M random relabelings, compute $T_1, \dots, T_M$, set $T_0 = T_{\text{obs}}$, and use

$$\hat{p} = \frac{1 + \sum_{i=1}^{M} I(T_i \geq T_{\text{obs}})}{M + 1}. \tag{3.2}$$

Proposition 3.4

Under H_0, $P(\hat{p} \leq \alpha) \leq \alpha$ for every M. Moreover $\hat{p}$ estimates the exact permutation p-value with standard error about $\sqrt{p(1-p)/M}$.

Proof. Given the multiset, $T_0, \dots, T_M$ are exchangeable under H_0 (the observed labeling is itself a uniform draw), so the rank of T_0 is uniform (ties broken against rejection) and $\hat{p} \leq j/(M+1)$ iff T_0 is among the j largest. Second claim: $\hat{p} \approx$ a mean of M Bernoulli(p) indicators. ■

The "+1" makes the Monte Carlo test exactly valid.

Random relabeling. Draw a uniform permutation of $1, \ldots, N$ and assign group 1 to its first m positions; each m-subset arises from $m!\,n!$ of the $N!$ permutations, so has probability $1/\binom{N}{m}$.

Alternatives and p-values. One-sided alternatives to $H_0 : F = G$ are stochastic orderings, $G(x) \le F(x)$ for all x or the reverse ($\Delta > 0$ or $\Delta < 0$ in a shift model). For $D = \bar{Y} - \bar{X}$ with values $d_0 = D_{\text{obs}}, d_1, \ldots, d_M$:

$$p_{\text{upper}} = \frac{\#\{d_i \ge d_0\}}{M+1}, \qquad p_{\text{lower}} = \frac{\#\{d_i \le d_0\}}{M+1}, \qquad p_{\text{two}} = \frac{\#\{|d_i| \ge |d_0|\}}{M+1}.$$

With $M = 1999$, the standard error at $p = 0.05$ is about 0.005.

3.5 Other permutation tests

One sample: sign flipping. If $X_i - \theta_0$ are independent and symmetric about 0 under H_0, all 2^n sign choices give the same distribution. Flipping signs with $\sum I(X_i > \theta_0)$ gives the sign test, with $\sum V_i R_i$ the signed-rank test, with the t-statistic a symmetric-F t-test.

2×2 tables: Fisher's exact test. For two binary variables and H_0: independence, permute one variable. With margins fixed, the (exposed, diseased) count D is hypergeometric:

$$P(D = d) = \binom{m}{d}\binom{N-m}{k-d}\Big/\binom{N}{k},$$

where m is the number diseased and k the number exposed among N.

Example 3.2

$N = 20$ subjects, $m = 10$ diseased, $k = 6$ exposed, and 5 of the exposed are diseased. Then $P(D \ge 5) = [\binom{10}{5}\binom{10}{1} + \binom{10}{6}\binom{10}{0}]/\binom{20}{6} = (2520 + 210)/38760 = 0.070$.

Summary. Permutation tests need only exchangeability under H_0, allow any statistic and give exact p-values; but they test only hypotheses making the data exchangeable (not "equal means, unequal variances"), and intervals require inverting many tests.

3.6 Selected exercises

Motivation. Monte Carlo error of $\hat{p}$, 2×2 statistics as Fisher tests, and moments of the permutation distribution.

Exercise 3.6 (Monte Carlo error of $\hat{p}$)

A random permutation test with $M = 999$ relabelings gave $\hat{p} = 0.01$. Give an approximate 99% confidence interval for the exact permutation p-value p. How much shorter would it be with $M = 1999$?

Solution.

Step 1. Given the data, each relabeling is uniform over the K splits, so $P(T_i \ge T_{\text{obs}}) = p$ by (3.1); independence gives $\sum_{i=1}^{M} I(T_i \ge T_{\text{obs}}) \sim B(M, p)$, and $\hat{p}$ in (3.2) differs from that proportion by $O(1/M)$. So $\hat{p} \dot{\sim} N(p, p(1-p)/M)$.

Step 2. With $z_{0.005} = 2.576$ and $\hat{p}$ in the standard error,

$$\hat{p} \pm 2.576\sqrt{\frac{\hat{p}(1-\hat{p})}{M}} = 0.01 \pm 2.576\sqrt{\frac{0.0099}{999}} = 0.01 \pm 0.0081 = (0.0019,\ 0.0181).$$

Step 3. With $M = 1999$ the half-width is $2.576\sqrt{0.0099/1999} = 0.0057$: $(0.0043,\ 0.0157)$. Width scales as $M^{-1/2}$, so doubling M shortens it by $1 - \sqrt{999/1999} \approx 29\%$; halving it needs $4M$. ■

Exercise 3.7–3.8 (χ^2 and odds ratio as permutation statistics)

In the 2×2 setting of Section 3.5 (margins fixed: N subjects, m diseased, k exposed, D = number exposed and diseased), show:
(a) Pearson's χ^2 statistic is an increasing function of $|D - km/N|$, so the permutation test based on χ^2 is the two-sided Fisher test that orders tables by $|D - ED|$;
(b) the sample odds ratio is an increasing function of D, so the permutation test based on it is the one-sided Fisher test.
Apply both to Example 3.2.

Solution. Cells: $a = D$, $b = k - D$, $c = m - D$, $d = N - m - k + D$; row totals $k, N-k$, column totals $m, N-m$.

(a)

Step 1. Expected counts are $E_a = km/N$, $E_b = k(N-m)/N$, $E_c = (N-k)m/N$, $E_d = (N-k)(N-m)/N$, so

$$a - E_a = d - E_d = D - \frac{km}{N}, \qquad b - E_b = c - E_c = -\Big(D - \frac{km}{N}\Big).$$

Step 2. Hence $\chi^2 = \sum(O-E)^2/E = (D - km/N)^2 \sum 1/E$, and

$$\sum \frac{1}{E} = \frac{N}{k\,m\,(N-k)(N-m)}\big[(N-k)(N-m)+(N-k)m+k(N-m)+km\big] = \frac{N^3}{k\,m\,(N-k)(N-m)}.$$

The constant is the same for every permutation (fixed margins), so χ^2 is increasing in $|D - km/N|$, and with $ED = km/N$ the p-value is $P(|D - km/N| \geq |d_{\text{obs}} - km/N|)$.

(b) Where $b, c > 0$, $\text{OR} = \frac{ad}{bc} = \frac{D}{m-D} \cdot \frac{N-m-k+D}{k-D}$, each factor nonnegative increasing over positive decreasing, so OR increases in D (strictly where positive). At the ends OR is 0 (a or $d = 0$) or $+\infty$ (b or $c = 0$). So "$\text{OR} \geq \text{OR}_{\text{obs}}$" is $D \geq d_{\text{obs}}$: the one-sided Fisher test.

In Example 3.2, $km/N = 3$, $d = 5$, and $m = N - m = 10$ makes $P(D = d)$ symmetric about 3. Odds ratio: $p = P(D \geq 5) = 0.070$; χ^2: $p = P(D \geq 5) + P(D \leq 1) = 0.141$. ■

Exercise 3.9 (Moments of the permutation distribution of $\bar{Y} - \bar{X}$)

Let $z_1, \ldots, z_N$ be the pooled observations, $S^2 = \frac{1}{N-1}\sum_k (z_k - \bar{z})^2$, and let $D = \bar{Y} - \bar{X}$ be computed for a uniformly random split into groups of sizes m (X) and n (Y).
(a) Show that $ED = 0$ and $\text{Var}\, D = NS^2/(mn)$.
(b) Let $Z = D/\sqrt{NS^2/(mn)}$ and let T be the pooled two-sample t-statistic. Show that $Z^2 = (N-1)T^2/(N-2+T^2)$. Hence Z is an increasing function of T, and $Z/T \to 1$ as $N \to \infty$ with T fixed.

Solution [Supplement].

(a) $\bar{Y}$ is the mean of an SRS of size n from $\{z_k\}$, population variance $\sigma^2 = (N-1)S^2/N$. By Theorem 4.2,

$$E\bar{Y} = \bar{z}, \qquad \operatorname{Var}\bar{Y} = \frac{\sigma^2}{n}\cdot\frac{N-n}{N-1} = \frac{mS^2}{nN}.$$

Since $m\bar{X} + n\bar{Y} = N\bar{z}$, $D = \frac{N}{m}(\bar{Y} - \bar{z})$, so $ED = 0$ and $\operatorname{Var} D = \frac{N^2}{m^2}\cdot\frac{mS^2}{nN} = \frac{NS^2}{mn}$.

(b) From the proof of Proposition 3.2, $(N-1)S^2 = SST = (N-2)s_p^2 + A$ with $A = \frac{mn}{N}D^2$, and $T^2 = A/s_p^2$. Therefore

$$Z^2 = \frac{A}{S^2} = \frac{(N-1)A}{(N-2)s_p^2 + A} = \frac{(N-1)T^2}{N-2+T^2}.$$

Both Z and T have the sign of D; thus $Z = g(T)$ with $g(t) = t\sqrt{(N-1)/(N-2+t^2)}$, where $g'(t) = \sqrt{N-1}\,(N-2)(N-2+t^2)^{-3/2} > 0$ and $g(t)/t \to 1$ as $N \to \infty$.

Hence permutation and normal-theory p-values nearly coincide in large samples. ■

Chapter 4. Inference for Two Populations

4.1 Models for two populations

$X_1, \dots, X_m \overset{\text{i.i.d.}}{\sim} F$ and $Y_1, \dots, Y_n \overset{\text{i.i.d.}}{\sim} G$ independent, $N = m + n$. Typical models:

- **Shift:** $G(x) = F(x - \Delta)$, i.e. $Y \overset{d}{=} X + \Delta$.
- **Scale:** $G(x) = F(x/\gamma)$ (common center 0), or location–scale $G(x) = F((x - \Delta)/\gamma)$.
- **General:** $H_0 : F = G$ against $F \neq G$.

Permutation tests need exchangeability under H_0, i.e. H_0 must force $F = G$ ($\Delta = 0$, $\gamma = 1$, or the general null); "equal locations, different scales" fails (Section 4.6).

For normal data: the pooled t-test (equal variances) or Welch's $T = (\bar{Y} - \bar{X})/\sqrt{s_X^2/m + s_Y^2/n}$ with Satterthwaite d.f.

4.2 The Wilcoxon rank-sum (Mann–Whitney) test

Motivation. Pooled ranks make the permutation distribution data-free, and the rank sum estimates $P(X < Y)$.

Rank all N observations together. Let $R_1, \dots, R_n$ be the ranks of the Y's.

Definition (Rank-sum and Mann–Whitney statistics)

$W = \sum_{j=1}^{n} R_j$ and $U = \sum_{i=1}^{m} \sum_{j=1}^{n} I(Y_j > X_i)$ (with $\frac{1}{2}$ for ties).

Proposition 4.1

Without ties, $W = U + n(n+1)/2$.

Proof. $R_j = 1 + \#\{k : Y_k < Y_j\} + \#\{i : X_i < Y_j\}$; summing over j the middle terms give $0 + \cdots + (n-1)$, so $W = n + \frac{n(n-1)}{2} + U$. ■

Under $H_0 : F = G$ (continuous) the Y-ranks form a uniform n-subset of $\{1, \dots, N\}$, so the null law of W depends only on (m, n): the permutation test on ranks. Large W indicates $\Delta > 0$.

Example 4.1

$X = \{1.1, 2.4\}$, $Y = \{2.0, 3.5, 4.2\}$: Y-ranks $2, 4, 5$, $W = 11$, $U = 5$. Of the 10 rank subsets only $\{2, 4, 5\}, \{3, 4, 5\}$ have sum ≥ 11: $P(W \geq 11) = 0.2$.

4.2.1 Large-sample approximation via sampling without replacement

Theorem 4.2

Let $\bar{X}_n$ be the mean of a simple random sample of size n drawn without replacement from a finite population $\{a_1, \dots, a_N\}$ with mean μ and variance $\sigma^2 = \frac{1}{N}\sum(a_k - \mu)^2$. Then

$$E\bar{X}_n = \mu, \qquad \operatorname{Var}\bar{X}_n = \frac{\sigma^2}{n} \cdot \frac{N-n}{N-1}.$$

Proof [Supplement]. Write $n\bar{X}_n = \sum_k Z_k a_k$ with inclusion indicators: $P(Z_k = 1) = n/N$, $P(Z_k = Z_l = 1) = \frac{n(n-1)}{N(N-1)}$ $(k \neq l)$, so $\operatorname{Var} Z_k = \frac{n}{N}(1 - \frac{n}{N})$, $\operatorname{Cov}(Z_k, Z_l) = -\frac{n(N-n)}{N^2(N-1)}$. With $\mu = 0$, $\sum_k a_k^2 = N\sigma^2$ and $\sum_{k \neq l} a_k a_l = -N\sigma^2$, so

$$\operatorname{Var}\Big(\sum Z_k a_k\Big) = N\sigma^2 \frac{n(N-n)}{N^2} + N\sigma^2 \frac{n(N-n)}{N^2(N-1)} = n\sigma^2 \frac{N-n}{N-1}.$$

■

$\frac{N-n}{N-1}$ is the **finite-population correction**. A CLT holds if $\min(n, N-n) \to \infty$ and $\max_k(a_k - \mu)^2/(\min(n, N-n)\sigma^2) \to 0$ (Hájek). For ranks, $\mu = \frac{N+1}{2}$, $\sigma^2 = \frac{N^2-1}{12}$:

$$E(W) = \frac{n(N+1)}{2}, \qquad \operatorname{Var}(W) = \frac{mn(N+1)}{12}, \qquad \frac{W - E(W)}{\sqrt{\operatorname{Var}(W)}} \xrightarrow{d} N(0,1).$$

With ties, use the mid-ranks as the population (mean unchanged):

$$\operatorname{Var}(W) = n^2 \cdot \frac{\sigma_R^2}{n} \cdot \frac{N-n}{N-1}, \qquad \sigma_R^2 = \text{population variance of the mid-ranks.} \tag{4.1}$$

Applied to the pooled observations it gives the large-sample permutation law of $\bar{Y}$, so permutation and normal-theory p-values agree for large samples.

4.2.2 Estimating the shift

In the shift model $U(\Delta_0) = \#\{(i,j) : Y_j - X_i > \Delta_0\}$ is the Mann–Whitney statistic of X and $Y - \Delta_0$; at the true Δ it has the null law, symmetric about $mn/2$.

Proposition 4.3

Let $D_{(1)} \le \cdots \le D_{(mn)}$ be the ordered differences $Y_j - X_i$, $k_l = Q_U(\alpha/2)$ and $k_u = mn - k_l + 1$. Then $P(D_{(k_l)} \le \Delta < D_{(k_u)}) = P(k_l \le U \le k_u - 1) \ge 1 - \alpha$. The associated point estimator is the Hodges–Lehmann shift estimator $\hat{\Delta} = \operatorname{median}_{i,j}(Y_j - X_i)$.

Proof. $D_{(k_l)} \le \Delta \iff U(\Delta) \le mn - k_l$ and $\Delta < D_{(k_u)} \iff U(\Delta) \ge mn - k_u + 1$; use the symmetric null law of $U(\Delta)$. ■

In Example 4.1 the differences are $-0.4, 0.9, 1.1, 1.8, 2.4, 3.1$, so $\hat{\Delta} = 1.45$. For large samples $Q_U(\alpha) \approx \frac{mn}{2} - z_\alpha\sqrt{mn(N+1)/12}$.

Ranks as a probability transform. Under H_0, $U_i = F(Z_i)$ are i.i.d. uniform and $E\,U_{(k)} = k/(N+1)$: up to $N+1$, ranks are expected uniform order statistics. The rank-sum test is a permutation test after implicitly transforming F to the uniform; hence its null law is the same for every continuous F, and it needs $F = G$ (a common transform) under H_0.

Paired data. Apply Chapter 2 to $Y_i - X_i$.

4.3 Tests for scale

Assume a common center and $G(x) = F(x/\gamma)$. A scale change does not shift W, so use *scores* large in the tails:

- **Ansari–Bradley:** scores $1, 2, 3, \ldots$ from both ends inward; a small Y-score sum means Y is more dispersed.
- **Siegel–Tukey:** $1, 2, \ldots, N$ alternating from both ends, so the null law equals that of W.
- **Mood:** scores $(R - \frac{N+1}{2})^2$; a large Y-score sum means Y is more dispersed.

Each is a permutation test of a score sum, distribution-free under $F = G$. If the centers may differ:

- Under normality, $F = S_Y^2/S_X^2 \sim F_{n-1,m-1}$ when $\gamma = 1$; very sensitive to non-normality and outliers.
- $\mathrm{MAD}(Y)/\mathrm{MAD}(X)$, a robust alternative.
- Score statistics after centering each sample at its median.

Then $F = G$ is not the null, so permute the median-centered (mean-centered for the F ratio) observations; this is only *approximately* valid. Without centering, a location difference masquerades as a scale difference.

4.4 The Kolmogorov–Smirnov test

For general $H_0 : F = G$ against $F \neq G$, compare the empirical distribution functions $\hat{F}_m(x) = \frac{1}{m}\sum I(X_i \le x)$ and $\hat{G}_n$:

$$K = \sqrt{\frac{mn}{m+n}}\,\sup_x |\hat{F}_m(x) - \hat{G}_n(x)|.$$

The supremum is attained at an observation, so K depends only on the ranks and is distribution-free (continuous F). As $m, n \to \infty$, $P(K \le x) \to 1 - 2\sum_{j\ge 1}(-1)^{j-1}e^{-2j^2x^2}$ (Kolmogorov). KS detects any difference but is less powerful than the rank-sum test against a pure shift.

Why K depends only on ranks [Supplement]. Both EDFs are constant between consecutive pooled order statistics, so the supremum is a maximum over the N pooled points; at the kth point $\hat{F}_m, \hat{G}_n$ are the proportions of X's and Y's among the k smallest, determined by the ranks, which are uniform over $\binom{N}{m}$ splits under H_0. ■

4.5 Which test?

With $F = G$ under H_0 every permutation or rank test is exact. The Pitman ARE of rank-sum vs. t is $12\sigma^2(\int f^2)^2$, as in one sample:

F	uniform	normal	logistic	Laplace	Cauchy
ARE(Wilcoxon, t)	1	$3/\pi \approx 0.955$	$\pi^2/9 \approx 1.10$	1.5	∞

(Exponential shift family: 3.) The two-sample t-test is close to nominal for light tails and conservative for heavy-tailed or skewed populations. In small-sample power studies the permutation t-test leads for uniform data, the t-test for normal, the rank-sum test for Laplace-type tails and the difference-of-medians permutation test for Cauchy-type tails.

4.6 Unequal variances under the null

If H_0 is "equal locations" with different scales, no rank or permutation test is guaranteed to hold its level. Use Welch's t-test, a variance-stabilizing transformation, a studentized rank statistic (Fligner–Policello, Chung–Romano), or the bootstrap tests of Chapter 9.

4.7 Selected exercises

Motivation. The probability integral transform, why the Mann–Whitney count is not binomial, and the Ansari–Bradley null moments.

Exercise 4.6 (Probability integral transform)

Let $X_1, \dots, X_N \overset{\text{i.i.d.}}{\sim} F$ with F continuous.
(a) Show that $U_i = F(X_i)$ are i.i.d. Uniform$(0,1)$.
(b) With $U_{(1)} < \cdots < U_{(N)}$ the order statistics, show that $E\,U_{(i)} = i/(N+1)$.

Solution.

(a)

Step 1. Fix $u \in (0,1)$ and let $x_u = \sup\{x : F(x) \le u\}$ (nonempty, bounded above since $F(-\infty) = 0$, $F(\infty) = 1$). $F(x) \le u$ for $x < x_u$ and $F(x) > u$ for $x > x_u$; continuity gives $F(x_u) = u$.

Step 2. So $\{F(X) \le u\} = \{X \le x_u\}$ and $P(F(X) \le u) = F(x_u) = u$.

Step 3. The U_i are functions of the independent X_i, hence independent.

(b)

Step 1. $U_{(i)} > u$ iff fewer than i of the U's are $\le u$, and $\#\{j : U_j \le u\} \sim B(N,u)$:

$$P(U_{(i)} > u) = \sum_{k=0}^{i-1} \binom{N}{k} u^k (1-u)^{N-k}.$$

Step 2. $EV = \int_0^\infty P(V > u)\,du$ for $V \ge 0$, and $\int_0^1 u^k(1-u)^{N-k}\,du = \frac{k!\,(N-k)!}{(N+1)!}$, so

$$E\,U_{(i)} = \sum_{k=0}^{i-1} \binom{N}{k} \frac{k!\,(N-k)!}{(N+1)!} = \sum_{k=0}^{i-1} \frac{1}{N+1} = \frac{i}{N+1}.$$

This justifies "Ranks as a probability transform" in Section 4.2.2. ■

Exercise 4.8 (Why $U(\Delta)$ is not binomial)

In the shift model, X_i and $Y_j - \Delta$ are independent with the same continuous distribution.
(a) Show that every indicator $I(Y_j - X_i > \Delta)$ is Bernoulli$(\frac{1}{2})$.
(b) Explain why $U(\Delta) = \sum_{i,j} I(Y_j - X_i > \Delta)$ is nevertheless not $B(mn, \frac{1}{2})$, as the analogous count was in the sign test. Compute $\operatorname{Var} U(\Delta)$ and compare it with $mn/4$.

Solution. Put $Y'_j = Y_j - \Delta$: $X_1, \dots, X_m, Y'_1, \dots, Y'_n$ are i.i.d. continuous, $A_{ij} = I(Y'_j > X_i)$.

(a) $P(X_i = Y'_j) = \int P(X_i = y)\, dF(y) = 0$, and by exchangeability $P(Y'_j > X_i) = P(X_i > Y'_j) = \frac{1}{2}$.

(b)

Step 1. Indicators sharing an index are dependent. For $j \neq j'$,

$$P(A_{ij} = A_{ij'} = 1) = P(X_i < \min(Y'_j, Y'_{j'})) = \frac{1}{3},$$

(each of three i.i.d. continuous variables is smallest with probability $\frac{1}{3}$), so $\operatorname{Cov}(A_{ij}, A_{ij'}) = \frac{1}{12}$; likewise $\operatorname{Cov}(A_{ij}, A_{i'j}) = P(\max(X_i, X_{i'}) < Y'_j) - \frac{1}{4} = \frac{1}{12}$. Indicators with no common index are independent.

Step 2. With $mn(n-1)$ and $mn(m-1)$ ordered pairs of the two kinds,

$$\operatorname{Var} U(\Delta) = \frac{mn}{4} + \frac{mn(n-1) + mn(m-1)}{12} = \frac{mn(N+1)}{12}.$$

Step 3. This exceeds $mn/4$ whenever $N \geq 3$, so binomial quantiles would give too-short intervals; it equals $\operatorname{Var}(W)$ (Section 4.2.1), as $U(\Delta)$ has the null law of $W - n(n+1)/2$. In the sign test the indicators involve different observations; here each X_i meets every Y'_j. ■

Exercise 4.11 (Ansari–Bradley moments)

Let $N = m+n$ be even. The Ansari–Bradley test assigns the scores $1, 2, \dots, N/2, N/2, \dots, 2, 1$ to the pooled order positions, and AB is the sum of the scores of the Y's. Find $E(AB)$ and $\operatorname{Var}(AB)$ under $H_0 : F = G$.

Solution.

Step 1. Under H_0 the Y-positions are a uniform n-subset, so AB is n times the mean of a sample without replacement from the scores, each of $1, \dots, N/2$ occurring twice.

Step 2. Population mean and mean square:

$$\mu = \frac{2}{N} \sum_{k=1}^{N/2} k = \frac{N+2}{4}, \qquad \frac{2}{N} \sum_{k=1}^{N/2} k^2 = \frac{(N+2)(N+1)}{12},$$

so $\sigma^2 = \frac{(N+2)(N+1)}{12} - \frac{(N+2)^2}{16} = \frac{(N+2)(N-2)}{48}$.

Step 3. By Theorem 4.2,

$$E(AB) = n\mu = \frac{n(N+2)}{4}, \qquad \operatorname{Var}(AB) = n^2 \cdot \frac{\sigma^2}{n} \cdot \frac{N-n}{N-1} = \frac{mn(N+2)(N-2)}{48(N-1)}.$$

Check: $N = 2$ gives $AB \equiv 1$, $E = 1$, $\operatorname{Var} = 0$. ■

Chapter 5. Association and Independence of Two Variables

5.1 Pearson correlation

$\rho = \mathrm{Cov}(X,Y)/\sqrt{\mathrm{Var}\,X\,\mathrm{Var}\,Y} \in [-1,1]$ measures *linear* association; $\rho = 0$ implies independence under bivariate normality only (e.g. $Y = X^2$, X symmetric, has $\rho = 0$). The sample correlation is $r = S_{XY}/\sqrt{S_{XX}S_{YY}}$ and the LS slope $\hat\beta = S_{XY}/S_{XX} = r\sqrt{S_{YY}/S_{XX}}$; since $\beta = \rho\,\sigma_Y/\sigma_X$, testing $\beta = 0$ and $\rho = 0$ coincide.

Proposition 5.1

If (X_i, Y_i) are i.i.d. bivariate normal with $\rho = 0$, then $T = r\sqrt{(n-2)/(1-r^2)} \sim t_{n-2}$.

Proof. Given the X's, the Y_i follow the regression model with $\beta = 0$, so $\hat\beta/\widehat{\mathrm{se}}(\hat\beta) \sim t_{n-2}$. With $SSE = S_{YY}(1-r^2)$ and $\widehat{\mathrm{se}}^2 = SSE/((n-2)S_{XX})$, $\hat\beta/\widehat{\mathrm{se}} = r\sqrt{(n-2)/(1-r^2)}$; the conditional law is free of the X's. ■

Permutation test of independence. Under $H_0 : X \perp Y$, shuffling the Y's among the fixed X's preserves the joint law, giving an exact test with any statistic, e.g. r (sensitive to linear association). H_0 is independence, not $\rho = 0$.

Proposition 5.2

Over all $n!$ permutations of $(Y_1, \ldots, Y_n)$ with the X's fixed, the permutation distribution of r has mean 0 and variance $1/(n-1)$, whatever the data.

Proof. Let $a_i = X_i - \bar X$, $b_i = Y_i - \bar Y$, π uniform: $r = \sum_i a_i b_{\pi(i)}/\sqrt{S_{XX}S_{YY}}$, $Eb_{\pi(i)} = 0$, $\mathrm{Var}\,b_{\pi(i)} = S_{YY}/n$, $\mathrm{Cov}(b_{\pi(i)}, b_{\pi(j)}) = -S_{YY}/(n(n-1))$ $(i \neq j)$. With $\sum_{i\neq j} a_i a_j = -S_{XX}$,

$$\mathrm{Var}\left(\sum a_i b_{\pi(i)}\right) = \frac{S_{XX}S_{YY}}{n} + \frac{S_{XX}S_{YY}}{n(n-1)} = \frac{S_{XX}S_{YY}}{n-1}.$$

■

With a finite-population CLT, $\sqrt{n-1}\,r \dot\sim N(0,1)$ under permutation: a distribution-free Proposition 5.1.

5.2 Spearman's rank correlation

$r_s = \mathrm{Corr}(R,S)$, the Pearson correlation of the ranks R_i, S_i. It measures *monotone* association, is invariant under increasing transformations and robust to outliers.

Proposition 5.3

Without ties, $r_s = 1 - \dfrac{6\sum_i (R_i - S_i)^2}{n(n^2-1)}$.

Proof. Both rank vectors have mean $\frac{n+1}{2}$ and $\sum(R_i - \bar R)^2 = \frac{n(n^2-1)}{12}$. Expanding $\sum(R_i - S_i)^2$ gives $\sum(R_i - \bar R)(S_i - \bar S) = \frac{n(n^2-1)}{12} - \frac{1}{2}\sum(R_i - S_i)^2$; divide by $\frac{n(n^2-1)}{12}$. ■

The permutation test with r_s tests independence against monotone association; Proposition 5.2 applies verbatim.

5.3 Kendall's tau

Motivation. Kendall's tau counts concordant minus discordant pairs: monotone association from ranks only.

A pair of observations $i < j$ is **concordant** if $(X_i - X_j)(Y_i - Y_j) > 0$ and **discordant** if < 0. Let $U_{ij} = \operatorname{sgn}[(X_i - X_j)(Y_i - Y_j)]$.

Definition (Kendall's tau)

Sample: $\tau_n = \binom{n}{2}^{-1} \sum_{i<j} U_{ij} = (N_c - N_d)/\binom{n}{2}$. Population: for independent copies $(X, Y), (X', Y')$, $\tau = P\{(X - X')(Y - Y') > 0\} - P\{(X - X')(Y - Y') < 0\}$.

τ_n is unbiased for τ; unlike r_s, it has a clear population counterpart.

Example 5.1

$x = (1, 2, 3, 4, 5)$, $y = (2, 1, 4, 3, 5)$: only pairs $(1, 2)$, $(3, 4)$ are discordant, so $\tau_n = (8-2)/10 = 0.6$. Rank differences $(-1, 1, -1, 1, 0)$ give $r_s = 1 - 6 \cdot 4/120 = 0.8$, equal to r (the data are their own ranks).

Proposition 5.4 (Normal case)

If (X, Y) is bivariate normal with correlation ρ, then $\tau = \frac{2}{\pi} \arcsin \rho$.

Proof [Supplement]. $(U, V) = (X - X', Y - Y')$ is centered normal with correlation ρ: $U = Z_1$, $V = \rho Z_1 + \sqrt{1 - \rho^2} Z_2$. In polar coordinates $(Z_1, Z_2) = R(\cos\Theta, \sin\Theta)$, Θ uniform, and with $\rho = \sin\phi$, $V = R\sin(\Theta + \phi)$. $\cos\Theta$ and $\sin(\Theta + \phi)$ share a sign on total angle $\pi + 2\phi$, so $P(UV > 0) = \frac{1}{2} + \frac{\phi}{\pi}$ and $\tau = \frac{2\phi}{\pi}$. ■

Hence $\hat{\rho} = \sin(\pi\tau_n/2)$ is a robust estimator of a normal correlation.

Proposition 5.5

If $X \perp Y$ with continuous margins, then $E\tau_n = 0$ and $\operatorname{Var}\tau_n = \dfrac{2(2n+5)}{9n(n-1)}$. Moreover $\tau_n / \sqrt{\operatorname{Var}\tau_n} \xrightarrow{d} N(0, 1)$.

Proof [Supplement]. $U_{ij} = \pm 1$ with probability $\frac{1}{2}$, $\operatorname{Var} U_{ij} = 1$; pairs with no common index are independent. For a shared index, $E(U_{12}U_{13}) = \left(E[\operatorname{sgn}(X_1 - X_2)\operatorname{sgn}(X_1 - X_3)]\right)^2 = \left(E(2V-1)^2\right)^2 = \frac{1}{9}$, V uniform. With $\binom{n}{2} \cdot 2(n-2)$ ordered pairs of pairs sharing one index,

$$\operatorname{Var}\Big(\sum_{i<j} U_{ij}\Big) = \binom{n}{2}\Big(1 + \frac{2(n-2)}{9}\Big) = \binom{n}{2}\frac{2n+5}{9},$$

divide by $\binom{n}{2}^2$. Asymptotic normality: Hoeffding's U-statistic theory. ■

Under independence the exact null law of τ_n is its permutation distribution (Kendall's tables without ties). Three ways to test independence:

(i) a (random) permutation test;

(ii) Kendall's exact null distribution;

(iii) the normal approximation of Proposition 5.5, good from $n \approx 10$ except in extreme tails. Ties give $U_{ij} = 0$; software reports $\tau_b = (N_c - N_d)/\sqrt{(N_0 - T_X)(N_0 - T_Y)}$, $N_0 = \binom{n}{2}$, T_X, T_Y the pairs tied in X or Y. With ties only the permutation test stays exact.

Which measure? Bivariate normal: the t-test of r. Heavy-tailed margins with linear dependence: r_s, τ_n are more powerful. Monotone nonlinear: r_s or τ_n. All three can be near zero under strong non-monotone dependence (e.g. $Y = \sin(3X) + \epsilon$).

5.4 Selected exercises

Motivation. Shuffling in regression, a dependent pair with $\rho = \tau = 0$, and Kendall's null law via inversions.

Exercise 5.2 (When shuffling is justified)

Let $Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$ with (X_i, ϵ_i) i.i.d., X_i independent of ϵ_i, $0 < \operatorname{Var} X_i < \infty$ and $\operatorname{Var} \epsilon_i < \infty$. Show that (X_1, Y_1) and (X_1, Y_2) have the same joint distribution if $\beta_1 = 0$, and different joint distributions if $\beta_1 \neq 0$.

Solution.

If $\beta_1 = 0$: $(X_1, Y_1) = (X_1, \beta_0 + \epsilon_1)$ and $(X_1, Y_2) = (X_1, \beta_0 + \epsilon_2)$, with X_1 independent of ϵ_1 and of ϵ_2, so both have the product law of X and $\beta_0 + \epsilon$.

If $\beta_1 \neq 0$: $\operatorname{Cov}(X_1, Y_1) = \beta_1 \operatorname{Var} X_1 \neq 0$ while $\operatorname{Cov}(X_1, Y_2) = 0$ ($X_1 \perp (X_2, \epsilon_2)$), so the laws differ.

So shuffling Y among fixed X's preserves the joint law exactly when $\beta_1 = 0$, the null of the permutation test. ■

Exercise 5.5 (Dependence with $\rho = \tau = 0$)

Let $X \sim \text{Uniform}(-1, 1)$ and $Y = X^2$. Show that X and Y are dependent but $\rho = 0$ and $\tau = 0$.

Solution [Supplement].

Dependence. $Y < \frac{1}{4}$ iff $|X| < \frac{1}{2}$, so $P(X > \frac{1}{2}, Y < \frac{1}{4}) = 0$, while $P(X > \frac{1}{2})P(Y < \frac{1}{4}) = \frac{1}{4} \cdot \frac{1}{2} \neq 0$.

Pearson. $EX = EX^3 = 0$ by symmetry, so $\operatorname{Cov}(X, Y) = EX^3 - EX\, EX^2 = 0$.

Kendall. For an independent copy (X', Y'),

$$(X - X')(Y - Y') = (X - X')(X^2 - X'^2) = (X - X')^2(X + X').$$

Since $P(X = X') = 0$, its sign is a.s. that of $X + X'$, continuous and symmetric about 0; so $\tau = 0$.

Hence $E\tau_n = 0$ as under independence: r, r_s, τ_n have little power against U-shaped dependence. ■

Exercise 5.6 (Kendall's null distribution via inversions)

Assume no ties. List the pairs in increasing order of X and let $\pi(k)$ be the rank of the Y-value of the kth pair, so π is a permutation of $\{1, \ldots, n\}$. Let N_d be the number of discordant pairs. (a) Show that $N_d = \#\{j < k : \pi(j) > \pi(k)\}$ and $\tau_n = 1 - 4N_d/(n(n-1))$.

(b) Let $I_k = \#\{j < k : \pi(j) > \pi(k)\}$. Show that if π is uniform on the $n!$ permutations, then $I_1, \dots, I_n$ are independent and I_k is uniform on $\{0, \dots, k-1\}$.
(c) Deduce $E\tau_n = 0$ and $\operatorname{Var}\tau_n = 2(2n+5)/(9n(n-1))$ under independence of X and Y.

Solution [Supplement].

(a) For $j < k$ in X-order, the pair is discordant iff $\pi(j) > \pi(k)$. Without ties $N_c + N_d = \binom{n}{2}$, so $\tau_n = (\binom{n}{2} - 2N_d)/\binom{n}{2} = 1 - 4N_d/(n(n-1))$.

(b)

Step 1. $\pi \mapsto (I_1, \dots, I_n)$ maps the $n!$ permutations into $\prod_{k=1}^{n}\{0, \dots, k-1\}$, also of size $n!$; it suffices to show injectivity.

Step 2. Recover π backwards: if $\pi(k+1), \dots, \pi(n)$ are known, so is $V_k = \{\pi(1), \dots, \pi(k)\}$, and exactly I_k elements of V_k exceed $\pi(k)$, so $\pi(k)$ is the $(k - I_k)$th smallest of V_k. Start at $k = n$, $V_n = \{1, \dots, n\}$.

Step 3. So the map is a bijection, and uniform π gives a uniform point of the product: the I_k are independent, I_k uniform on $\{0, \dots, k-1\}$.

(c)

Step 1. Under independence the Y's of the X-ordered pairs are i.i.d. continuous given the X's, so π is uniform.

Step 2. $N_d = \sum_k I_k$, $EI_k = \frac{k-1}{2}$, $\operatorname{Var} I_k = \frac{k^2-1}{12}$, so $EN_d = \frac{n(n-1)}{4}$, $E\tau_n = 0$, and

$$\operatorname{Var} N_d = \frac{1}{12}\sum_{k=1}^{n}(k^2 - 1) = \frac{1}{12}\Big(\frac{n(n+1)(2n+1)}{6} - n\Big) = \frac{n(n-1)(2n+5)}{72}.$$

Step 3. Hence $\operatorname{Var}\tau_n = \frac{16}{n^2(n-1)^2} \cdot \frac{n(n-1)(2n+5)}{72} = \frac{2(2n+5)}{9n(n-1)}$, as in Proposition 5.5.

The exact null law follows from

$$Eq^{N_d} = \prod_{k=1}^{n} \frac{1 + q + \cdots + q^{k-1}}{k}.$$

and asymptotic normality from Lyapunov: $\sum_k E|I_k - EI_k|^3 \le \sum_k k^3 = O(n^4)$ while $(\operatorname{Var} N_d)^{3/2} \asymp n^{9/2}$. ∎

Chapter 6. Permutation Tests for Analysis of Variance

6.1 One-way layout

Motivation. The Kruskal–Wallis test compares the average pooled ranks of k groups, a rank version of one-way ANOVA.

Observations $X_{ij} = \mu_i + \epsilon_{ij}$, $i = 1, \dots, k$, $j = 1, \dots, n_i$, $n = \sum n_i$, with i.i.d. errors. We test $H_0 : \mu_1 = \cdots = \mu_k$.

The F-test. With group means $\bar{X}_i$ and grand mean $\bar{X}$,

$$SS_{\text{Tr}} = \sum_i n_i(\bar{X}_i - \bar{X})^2, \qquad SS_E = \sum_{i,j}(X_{ij} - \bar{X}_i)^2, \qquad F = \frac{SS_{\text{Tr}}/(k-1)}{SS_E/(n-k)}.$$

Proposition 6.1

$SS_{\text{Tot}} := \sum_{i,j}(X_{ij} - \bar{X})^2 = SS_{\text{Tr}} + SS_E$. Under H_0 with normal errors, $F \sim F_{k-1,n-k}$.

Proof [Supplement]. Expand $X_{ij} - \bar{X} = (X_{ij} - \bar{X}_i) + (\bar{X}_i - \bar{X})$; the cross term $2\sum_i(\bar{X}_i - \bar{X})\sum_j(X_{ij} - \bar{X}_i) = 0$. Under normality $SS_E/\sigma^2 \sim \chi^2_{n-k}$ independently of the group means, and SS_{Tr}/σ^2 is the squared projection of the standard normal $(\sqrt{n_i}(\bar{X}_i - \mu)/\sigma)_i$ onto the complement of $(\sqrt{n_i})_i$, so χ^2_{k-1}. ■

Permutation F-test. Under H_0 (equal distributions) the n observations are exchangeable: relabel among the $n!/\prod n_i!$ equally likely splits. With SS_{Tot} fixed, $F = \frac{n-k}{k-1} \cdot \frac{SS_{\text{Tr}}}{SS_{\text{Tot}} - SS_{\text{Tr}}}$ is increasing in SS_{Tr}. For moderate samples the permutation law of F is close to $F_{k-1,n-k}$ without normality.

6.1.1 The Kruskal–Wallis test

Rank jointly; $\bar{R}_i$ is the mean rank of group i, S_R^2 the sample variance of all ranks ($n(n+1)/12$ without ties).

Definition (Kruskal–Wallis statistic)

$KW = \frac{1}{S_R^2}\sum_{i=1}^k n_i\Big(\bar{R}_i - \frac{n+1}{2}\Big)^2$, which without ties equals $\frac{12}{n(n+1)}\sum_i n_i(\bar{R}_i - \frac{n+1}{2})^2$.

Proposition 6.2

(i) The permutation test based on KW coincides with the permutation F-test applied to the ranks.
(ii) Under H_0, $KW \xrightarrow{d} \chi^2_{k-1}$ as all $n_i \to \infty$.

Proof. (i) For ranks, $SS^R_{\text{Tot}} = (n-1)S_R^2$ is fixed and $SS^R_{\text{Tr}} = S_R^2\, KW$, so $F_R = \frac{n-k}{k-1}\frac{KW}{n-1-KW}$ is increasing in KW.

(ii) With $Z_i = \sqrt{n_i}(\bar{R}_i - \bar{R})/S_R$, Theorem 4.2 gives $\text{Cov}(Z) \to I - \mathbf{u}\mathbf{u}'$, $\mathbf{u} = (\sqrt{n_i/n})_i$, a rank-$(k-1)$ projection; a multivariate finite-population CLT gives $Z \xrightarrow{d} N(0, I - \mathbf{u}\mathbf{u}')$, so $KW = \|Z\|^2 \xrightarrow{d} \chi^2_{k-1}$. ■

In small samples prefer the exact or Monte Carlo permutation law to χ^2.

6.1.2 Ordered alternatives

Against $H_1 : \mu_1 \le \mu_2 \le \cdots \le \mu_k$ (at least one strict), the **Jonckheere–Terpstra** statistic sums the pairwise Mann–Whitney counts:

$$JT = \sum_{i<l} U_{il}, \qquad U_{il} = \#\{(a,b) : X_{ia} < X_{lb}\}.$$

Large JT supports the trend. Up to an affine map it is Kendall's τ between observations and group index; its permutation law gives an exact test, more powerful than KW against monotone trends.

6.2 Randomized complete block design

k treatments in each of b blocks, one observation per cell: $X_{ij} = \mu + t_i + \beta_j + \epsilon_{ij}$. We test $H_0 : t_1 = \cdots = t_k$.

Classical F. $SS_{\text{Tr}} = b\sum_i(\bar X_{i\cdot} - \bar X)^2$ and $SS_E = \sum_{i,j}(X_{ij} - \bar X_{i\cdot} - \bar X_{\cdot j} + \bar X)^2$. Under normality, $F = \frac{SS_{\text{Tr}}/(k-1)}{SS_E/((k-1)(b-1))} \sim F_{k-1,(k-1)(b-1)}$.

Why shuffle within blocks. Under H_0, $X_{ij} = \mu+\beta_j+\epsilon_{ij}$: observations are exchangeable within, not across, blocks. So permute labels independently within blocks, $(k!)^b$ arrangements. These leave block means and SS_{Tot} unchanged, so $SS_E = SS_{\text{Tot}} - SS_{\text{Blocks}} - SS_{\text{Tr}}$ and the permutation F-test is the permutation test of SS_{Tr}.

6.2.1 The Friedman test

Rank within blocks; $R_i = \sum_j R_{ij}$, $\bar R_{i\cdot} = R_i/b$.

Definition (Friedman statistic)

$$FM = \frac{12b}{k(k+1)} \sum_{i=1}^k \Big(\bar R_{i\cdot} - \frac{k+1}{2}\Big)^2 = \frac{12}{bk(k+1)} \sum_i R_i^2 - 3b(k+1) \text{ (no ties).}$$

Under H_0 the within-block rank vectors are independent uniform permutations, so the null law is distribution-free and $FM \xrightarrow{d} \chi^2_{k-1}$ as $b \to \infty$. With ties divide by the average within-block rank variance instead of $k(k+1)/12$.

Example 6.1

$k = 3$, $b = 4$, ranks of (A, B, C): $(1,2,3)$, $(1,3,2)$, $(1,2,3)$, $(2,1,3)$. Rank sums $5, 8, 11$ give $FM = \frac{12}{48}(25 + 64 + 121) - 48 = 4.5$: χ^2_2 gives $p = e^{-2.25} = 0.105$, exact enumeration of $(3!)^4 = 1296$ arrangements $p = 162/1296 = 0.125$.

6.2.2 Page's trend test

For $H_1 : t_1 \le \cdots \le t_k$ use $L = \sum_{i=1}^k iR_i$, a block sum of Spearman-type statistics.

Proposition 6.3

Under H_0, $E(L) = \dfrac{bk(k+1)^2}{4}$ and $\mathrm{Var}(L) = \dfrac{bk^2(k+1)(k^2-1)}{144}$.

Proof. In block j, $L_j = \sum_i iR_{ij}$ with a uniform rank permutation, $ER_{ij} = \frac{k+1}{2}$, so $EL_j = \frac{k+1}{2} \cdot \frac{k(k+1)}{2}$. Proposition 5.2 with $a_i = i$ gives $\mathrm{Var}\, L_j = \frac{1}{k-1}\Big(\sum_i (i - \frac{k+1}{2})^2\Big)^2 = \frac{1}{k-1}\Big(\frac{k(k^2-1)}{12}\Big)^2 = \frac{k^2(k+1)(k^2-1)}{144}$. Blocks are independent; multiply by b. ■

6.3 General two-way layouts

With c replicates per cell and no interaction, $X_{ijl} = \mu + t_i + \beta_j + \epsilon_{ijl}$; permute all kc observations within blocks. The rank version (**Skillings–Mack**) uses

$$SM = \frac{b\sum_i (\bar{R}_{i\cdot} - \bar{R})^2}{\text{average within-block rank variance}} \dot{\sim} \chi^2_{k-1}.$$

Prefer the permutation law in small designs. With interaction, ranking can destroy or create it: use aligned-rank procedures or the regression resampling of Chapter 10.

6.4 Selected exercises

Motivation. Friedman as the sign test and as Cochran's Q, and Page's moments with ties.

Exercise 6.5 (Friedman with two treatments is the sign test)

In a complete block design with $k = 2$ treatments, b blocks and no ties within blocks, show that the permutation test based on the Friedman statistic FM is the two-sided sign test applied to the differences $D_j = X_{1j} - X_{2j}$.

Solution.

Step 1. Let $S = \#\{j : D_j > 0\}$. Within block j treatment 1 has rank $R_{1j} = 1 + I(D_j > 0)$, so $R_1 = b + S$ and $R_2 = 3b - R_1 = 2b - S$.

Step 2. By the second form of the definition with $k = 2$,

$$\begin{aligned} FM &= \frac{2}{b}\{(b+S)^2 + (2b-S)^2\} - 9b = \frac{2}{b}(5b^2 - 2bS + 2S^2) - 9b \\ &= \frac{(2S-b)^2}{b} = \Big(\frac{S - b/2}{\sqrt{b/4}}\Big)^2. \end{aligned}$$

So FM is the squared standardized sign statistic, increasing in $|S - b/2|$.

Step 3. Within-block permutations swap the labels independently with probability $\frac{1}{2}$, flipping $\operatorname{sgn} D_j$; so $S \sim B(b, \frac{1}{2})$ (Proposition 2.1), and large FM means large $|S - b/2|$: the two-sided sign test, with χ^2_1 as its normal approximation.

So for $k = 2$ and normal errors its ARE relative to the paired t-test is $2/\pi$ (Section 2.3); the general value $3k/(\pi(k+1))$ (not proved here) increases toward $3/\pi$. ■

Exercise 6.6 (Page's L with ties)

In a complete block design rank within blocks using mid-ranks, and let S_j^2 be the sample variance (divisor $k-1$) of the ranks in block j. Show that under H_0

$$E(L) = \frac{bk(k+1)^2}{4}, \qquad \mathrm{Var}(L) = \frac{k(k-1)(k+1)}{12}\sum_{j=1}^{b} S_j^2,$$

and that this reduces to Proposition 6.3 without ties.

Solution.

Step 1. Under H_0, given the multiset of mid-ranks in block j, $(R_{1j}, \ldots, R_{kj})$ is a uniform permutation of it; blocks are independent.

Step 2. Mid-ranks keep the total $k(k+1)/2$, so $ER_{ij} = \frac{k+1}{2}$ and $E(L_j) = \frac{k+1}{2}\sum_i i = \frac{k(k+1)^2}{4}$, $L_j = \sum_i i R_{ij}$.

Step 3. The proof of Proposition 5.2 needs only fixed $b_1, \ldots, b_k$ permuted against fixed $a_1, \ldots, a_k$: $\mathrm{Var}\left(\sum_i a_i b_{\pi(i)}\right) = S_{aa}S_{bb}/(k-1)$, $S_{aa} = \sum_i (a_i - \bar{a})^2$. With $a_i = i$ ($S_{aa} = k(k^2-1)/12$) and the mid-ranks of block j ($S_{bb} = (k-1)S_j^2$),

$$\mathrm{Var}(L_j) = \frac{k(k^2-1)}{12} S_j^2.$$

Step 4. Sum over independent blocks. Without ties $S_j^2 = k(k+1)/12$ and $\mathrm{Var}(L) = \frac{k(k-1)(k+1)}{12} \cdot \frac{bk(k+1)}{12} = \frac{bk^2(k+1)(k^2-1)}{144}$. ■

Exercise 6.7 (Binary responses: Cochran's Q)

Let $x_{ij} \in \{0, 1\}$ (success or failure of treatment i in block j), with treatment totals T_i, block totals B_j and grand total T. The tie-corrected Friedman statistic is $FM = b\sum_i (\bar{R}_{i\cdot} - \frac{k+1}{2})^2 / (\frac{1}{b}\sum_j S_j^2)$, with mid-ranks and S_j^2 as in Exercise 6.6.
(a) Show that FM equals Cochran's statistic

$$Q = \frac{(k-1)(k\sum_i T_i^2 - T^2)}{kT - \sum_j B_j^2}.$$

(b) Four treatments of plantar warts were applied to each of 7 patients (blocks), with outcomes (1 = success):

	P1	P2	P3	P4	P5	P6	P7
Freezing	0	1	0	0	0	1	0
Surgery	1	1	0	1	0	1	1
Topical medication	0	0	0	1	0	1	0
Duct tape	1	1	0	0	1	0	1

Test whether the treatments differ.

Solution.

(a)

Step 1. (Mid-ranks.) In block j the $k - B_j$ failures get $c_j = \frac{k - B_j + 1}{2}$ and the successes $\frac{2k - B_j + 1}{2} = c_j + \frac{k}{2}$, so $R_{ij} = c_j + \frac{k}{2}x_{ij}$ (also for $B_j \in \{0, k\}$).

Step 2. (Numerator.) $\bar R_{i\cdot} = \bar c + \frac{k}{2b}T_i$, whose average over i is $\frac{k+1}{2}$, so $\bar R_{i\cdot} - \frac{k+1}{2} = \frac{k}{2b}(T_i - T/k)$ and

$$b\sum_i \Big(\bar R_{i\cdot} - \frac{k+1}{2}\Big)^2 = \frac{k^2}{4b}\Big(\sum_i T_i^2 - \frac{T^2}{k}\Big).$$

Step 3. (Denominator.) $S_j^2 = \frac{k^2}{4}\cdot\frac{1}{k-1}\sum_i\big(x_{ij} - \frac{B_j}{k}\big)^2 = \frac{k^2}{4(k-1)}\Big(B_j - \frac{B_j^2}{k}\Big) = \frac{kB_j(k-B_j)}{4(k-1)}$.

Step 4. (Ratio.) Using $\sum_j B_j(k-B_j) = kT - \sum_j B_j^2$,

$$FM = \frac{\frac{k^2}{4b}\big(\sum_i T_i^2 - T^2/k\big)}{\frac{1}{b}\cdot\frac{k}{4(k-1)}\big(kT - \sum_j B_j^2\big)} = \frac{(k-1)\big(k\sum_i T_i^2 - T^2\big)}{kT - \sum_j B_j^2} = Q.$$

Blocks with $B_j \in \{0, k\}$ contribute to neither part: they carry no information.

(b) $T_i = (2,5,2,4)$, $T = 13$, $\sum T_i^2 = 49$; $B_j = (2,3,0,2,1,3,2)$, $\sum B_j^2 = 31$. So

$$Q = \frac{3(4\cdot 49 - 169)}{4\cdot 13 - 31} = \frac{81}{21} = 3.86.$$

The χ^2_3 approximation gives $p = 0.28$, while enumerating all 13,824 within-block arrangements gives the exact $p = 5148/13824 = 0.37$. No evidence of a difference; with six informative blocks χ^2 is noticeably off, as in Example 6.1. ■

Chapter 7. The Bootstrap

7.1 The plug-in principle

Let $X_1, \dots, X_n \overset{\text{i.i.d.}}{\sim} F$ and $\hat{\theta} = T(X_1, \dots, X_n)$ estimate $\theta = \theta(F)$. The **sampling distribution** G_F of $\hat{\theta}$ (bias, standard error, quantiles) depends on the unknown F. The bootstrap replaces F by an estimate $\hat{F}$ and simulates:

$$\text{real world: } F \to \mathbb{X}_n \to \hat{\theta}, \qquad \text{bootstrap world: } \hat{F} \to \mathbb{X}_n^* \to \hat{\theta}^*.$$

The distribution of $\hat{\theta}^*$ given the data, $G^* = G_{\hat{F}}$, is the **bootstrap distribution**; it estimates G_F.

Nonparametric bootstrap. $\hat{F} = F_n$ (mass $1/n$ at each observation): draw n observations *with replacement*.

1. Draw a bootstrap sample $X_1^*, \dots, X_n^*$ with replacement from $\{X_1, \dots, X_n\}$ and compute $\hat{\theta}^* = T(X_1^*, \dots, X_n^*)$.
2. Repeat B times to get $\hat{\theta}^{*1}, \dots, \hat{\theta}^{*B}$; their empirical distribution approximates G^*.

Two errors: $F_n \neq F$ (statistical, $O(n^{-1/2})$) and $B < \infty$ (Monte Carlo, $O(B^{-1/2})$, reducible at will).

7.2 Bootstrap estimates of standard error and bias

Replacing F by F_n in $\text{se}_F(\hat{\theta})$ and $\text{bias}_F(\hat{\theta}) = E_F\hat{\theta} - \theta(F)$ gives

$$\widehat{\text{se}}_{\text{Boot}} = \Big(\frac{1}{B-1}\sum_b (\hat{\theta}^{*b} - \bar{\theta}^*)^2\Big)^{1/2}, \qquad \widehat{\text{bias}}_{\text{Boot}} = \bar{\theta}^* - \hat{\theta}, \qquad \bar{\theta}^* = \frac{1}{B}\sum_b \hat{\theta}^{*b}.$$

The true $\theta(F)$ is replaced by $\theta(F_n)$, which is $\hat{\theta}$ for plug-in statistics.

Example 7.1 (An exact bootstrap distribution)

Data $\{1, 4, 9\}$, $\hat{\theta} = \text{median} = 4$. Of the 27 equally likely resamples, the median is 1 iff at least two draws are 1: probability $(3 \cdot 2 + 1)/27 = 7/27$; likewise 9 w.p. $7/27$ and 4 w.p. $13/27$. Hence

$$E^*\hat{\theta}^* = \frac{7 \cdot 1 + 13 \cdot 4 + 7 \cdot 9}{27} = 4.52, \qquad \text{se}^* = \Big(\frac{7 + 13 \cdot 16 + 7 \cdot 81}{27} - 4.52^2\Big)^{1/2} = 2.92,$$

and the bias estimate is 0.52, computed exactly since n is tiny.

When the bootstrap works

For smooth statistics (smooth functions of means, many M- and U-statistics) the bootstrap law of $\sqrt{n}(\hat{\theta}^* - \hat{\theta})$ consistently estimates that of $\sqrt{n}(\hat{\theta} - \theta)$. It can fail at the boundary: for $\hat{\theta} = \max_i X_i$, Uniform$(0, \theta)$, $P^*(\hat{\theta}^* = \hat{\theta}) = 1 - (1 - 1/n)^n \to 1 - e^{-1}$ but $P(\hat{\theta} = \theta) = 0$.

7.3 Parametric bootstrap

If $F = F_\eta$ is parametric, estimate $\hat{\eta}$ (e.g. by ML) and resample from $F_{\hat{\eta}}$. More accurate when the model is right and useful for intractable sampling laws (e.g. nonlinear functionals of a Weibull fit); badly biased when it is wrong.

7.4 Resampling designs

Rule: *resample the way the data were generated.*

- **Several samples.** Resample within each sample, keeping $n_1, \ldots, n_k$.
- **Random design.** If $(X_i, Y_i) \overset{\text{i.i.d.}}{\sim} F$, resample pairs (**case** bootstrap); valid even if the regression model is wrong.
- **Fixed design.** If $Y_i = \alpha + \beta x_i + \epsilon_i$ with i.i.d. errors, set $Y_i^* = \hat{\alpha} + \hat{\beta} x_i + r_i^*$, r_i^* drawn from the (leverage-adjusted) residuals; more efficient but relies on the model, including homoscedasticity.

Choosing:

- Classical inference is conditional on the x's; the residual bootstrap reproduces this and matches normal theory when it is adequate.
- The case bootstrap stays valid for a wrong linear model (β = minimizer of $E(Y - \alpha - \beta X)^2$) and heteroscedasticity, at some loss of efficiency.
- Designed experiments (x's chosen) argue for fixed x.

The bootstrap samples *with* replacement and keeps the structure of the alternative (pairs stay together); a permutation test samples *without* replacement and destroys the association to produce the null.

7.5 Selected exercises

Motivation. Ideal bootstrap ($B = \infty$) computations by hand, the uniform-maximum failure, and counting resamples.

Exercise 7.1 (The empirical distribution function)

Let $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} F$ and $F_n(x) = n^{-1} \sum_i I(X_i \le x)$. For x with $F(x) \in (0, 1)$, show:
(a) $nF_n(x) \sim B(n, F(x))$;
(b) $F_n(x)$ is unbiased for $F(x)$ with standard error $\sqrt{F(x)(1 - F(x))/n}$.

Solution.

(a) The $I(X_i \le x)$ are independent Bernoulli$(F(x))$, so $nF_n(x) \sim B(n, F(x))$.

(b) From (a), $E\, nF_n(x) = nF(x)$ and Var $nF_n(x) = nF(x)(1 - F(x))$. Divide by n and n^2.

So F_n is within $O_P(n^{-1/2})$ of F at each x: the statistical error of Section 7.1. ■

Exercise 7.5 (Ideal bootstrap for the mean and the variance)

Let $x_1, \dots, x_n$ be fixed data with mean $\bar{x}$, $\hat{\sigma}^2 = \frac{1}{n}\sum_i (x_i - \bar{x})^2$ and $s^2 = \frac{n}{n-1}\hat{\sigma}^2$, and let $X_1^*, \dots, X_n^*$ be drawn with replacement.
(a) Show that $E^*\bar{X}^* = \bar{x}$ and $\mathrm{Var}^*\,\bar{X}^* = \hat{\sigma}^2/n$.
(b) For $\theta = \mu^2$ and $\hat{\theta}_1 = \bar{X}^2$, show that the ideal bootstrap bias estimate is $\hat{\sigma}^2/n$. Compare it with the true bias, and find the bias of the corrected estimator $2\hat{\theta}_1 - \bar{\theta}^*$.
(c) For $\theta = \sigma^2$ and $\hat{\theta}_2 = s^2$, show that $\bar{\theta}^* - \hat{\theta}_2 = -s^2/n$, although s^2 is unbiased. Explain.

Solution [Supplement].

(a) The X_i^* are i.i.d. F_n: $E^*X_1^* = \bar{x}$, $\mathrm{Var}^*\,X_1^* = \hat{\sigma}^2$. Average n copies.

(b) $E^*(\bar{X}^{*2}) = \mathrm{Var}^*\,\bar{X}^* + (E^*\bar{X}^*)^2 = \hat{\sigma}^2/n + \bar{x}^2$, so $\bar{\theta}^* - \hat{\theta}_1 = \hat{\sigma}^2/n$; the true bias is $\mathrm{Var}\,\bar{X} = \sigma^2/n$. The corrected estimator is $\bar{X}^2 - \hat{\sigma}^2/n$, and with $E\hat{\sigma}^2 = \frac{n-1}{n}\sigma^2$,

$$E(\bar{X}^2 - \hat{\sigma}^2/n) = \mu^2 + \frac{\sigma^2}{n} - \frac{(n-1)\sigma^2}{n^2} = \mu^2 + \frac{\sigma^2}{n^2}.$$

The bias drops from $O(n^{-1})$ to $O(n^{-2})$.

(c) s^{*2} is unbiased for the variance $\hat{\sigma}^2$ of F_n, so $\bar{\theta}^* = \hat{\sigma}^2$ and $\bar{\theta}^* - s^2 = -s^2/n$. The bootstrap bias is $E_{F_n}\hat{\theta}^* - \theta(F_n)$ with $\theta(F_n) = \hat{\sigma}^2 \neq s^2$; correctly centered it is 0. The shortcut $\bar{\theta}^* - \hat{\theta}$ holds only for plug-in statistics (Section 7.2). ■

Exercise 7.6 (Failure for the uniform maximum)

Let $X_1, \dots, X_n \overset{\text{i.i.d.}}{\sim} \mathrm{Uniform}(0, \theta)$ and $\hat{\theta} = \max_i X_i$.
(a) Show that $n(\theta - \hat{\theta})/\theta \xrightarrow{d} \mathrm{Exp}(1)$.
(b) Show that $P^*(\hat{\theta}^* = \hat{\theta}) = 1 - (1 - 1/n)^n \to 1 - e^{-1}$.
(c) Conclude that the bootstrap distribution of $n(\hat{\theta} - \hat{\theta}^*)/\hat{\theta}$ does not converge to $\mathrm{Exp}(1)$.

Solution [Supplement].

(a) For $t \geq 0$ and $n > t$, $P(n(\theta - \hat{\theta})/\theta > t) = P(\hat{\theta} < \theta(1 - t/n)) = (1 - t/n)^n \to e^{-t}$.

(b) A.s. $x_{(n)}$ is a single point, and $\hat{\theta}^* = \hat{\theta}$ iff it is drawn at least once in n draws of probability $1/n$: $1 - (1 - 1/n)^n \to 1 - e^{-1} \approx 0.632$.

(c) Let $V^* = n(\hat{\theta} - \hat{\theta}^*)/\hat{\theta} \geq 0$. Convergence to the continuous $\mathrm{Exp}(1)$ would give $P^*(V^* \leq \frac{1}{2}) \to 1 - e^{-1/2} = 0.393$, but $P^*(V^* \leq \frac{1}{2}) \geq P^*(V^* = 0) \to 0.632$: the bootstrap law keeps an atom at 0. The cause: $\hat{\theta}$ sits at the boundary of the support, where F_n cannot mimic F. ■

Exercise 7.7 (Counting bootstrap samples)

Let the data $x_1, \dots, x_n$ be distinct.
(a) Show that there are $\binom{2n-1}{n}$ distinct bootstrap samples (as multisets), and that the one in which x_i appears c_i times has probability $\frac{n!}{c_1! \cdots c_n!} n^{-n}$.
(b) Show that the expected number of distinct original observations in a bootstrap sample is $n\{1 - (1 - 1/n)^n\} \approx 0.632\, n$.

Solution [Supplement].

(a) A multiset is given by counts $c_i \geq 0$, $\sum c_i = n$: n stars and $n - 1$ bars, $\binom{2n-1}{n}$ arrangements. The n^n ordered draws are equally likely, $n!/(c_1! \cdots c_n!)$ of them with given counts.

(b) With $J_i = I(x_i \text{ appears})$, $EJ_i = 1 - (1 - 1/n)^n$; sum over i, and $(1 - 1/n)^n \to e^{-1}$.

For $n = 3$: 10 multisets among 27 draws (Example 7.1), e.g. $\{1, 4, 9\}$ w.p. $6/27$, $\{1, 1, 1\}$ w.p. $1/27$. Since $\binom{2n-1}{n} \approx 4^n/(2\sqrt{\pi n})$, enumeration is infeasible and Monte Carlo is used. ■

Chapter 8. Bootstrap Confidence Intervals

Setting

From replicates $\hat\theta^{*1},\dots,\hat\theta^{*B}$ (law G^*, quantiles Q_{G^*}) we build a $1-\alpha$ interval; the methods differ in what they treat as a pivot.

8.1 Normal and basic intervals

Normal-approximation bootstrap interval. If $(\hat\theta-\text{bias}-\theta)/\,\text{se}\,\dot\sim N(0,1)$,

$$\theta \in \hat\theta - \widehat{\text{bias}}_{\text{Boot}} \pm z_{\alpha/2}\widehat{\text{se}}_{\text{Boot}}.$$

It uses two summaries of G^* and is as good as the normal approximation.

Basic interval. Assume $\hat\theta-\theta\sim H$ free of θ, approximated by the law of $\hat\theta^*-\hat\theta$ (quantiles $Q_{G^*}(p)-\hat\theta$). From $P(Q_H(\frac{\alpha}{2})\le\hat\theta-\theta\le Q_H(1-\frac{\alpha}{2}))=1-\alpha$,

$$\theta\in\big(2\hat\theta-Q_{G^*}(1-\tfrac{\alpha}{2}),\ 2\hat\theta-Q_{G^*}(\tfrac{\alpha}{2})\big). \tag{8.1}$$

8.2 The bootstrap-t interval

Assume $t=(\hat\theta-\theta)/\widehat{\text{se}}$ is approximately pivotal, approximated by

$$t^*=\frac{\hat\theta^*-\hat\theta}{\widehat{\text{se}}^*},$$

with $\widehat{\text{se}}^*$ recomputed on each resample. With Q_{t^*} the quantiles of the t^{*b},

$$\theta\in\big(\hat\theta-\widehat{\text{se}}\,Q_{t^*}(1-\tfrac{\alpha}{2}),\ \hat\theta-\widehat{\text{se}}\,Q_{t^*}(\tfrac{\alpha}{2})\big). \tag{8.2}$$

Cheap if $\widehat{\text{se}}$ has a formula ($s/\sqrt{n}$, regression SEs); otherwise a second-level bootstrap ($B\times C$, $C\approx 25$). Use a scale where t is nearly pivotal, e.g. $\log\hat\theta$ for a variance, then back-transform (coverage is preserved).

8.3 Percentile intervals

Motivation. Quantiles of the bootstrap distribution give an interval without a standard error formula.

Percentile interval. $\theta\in(Q_{G^*}(\frac{\alpha}{2}),Q_{G^*}(1-\frac{\alpha}{2}))$.

Proposition 8.1

Suppose there is an increasing φ, not necessarily known, with $\varphi(\hat\theta)-\varphi(\theta)\sim N(0,1)$, and the same holds in the bootstrap world: $\varphi(\hat\theta^*)-\varphi(\hat\theta)\sim N(0,1)$ given the data. Then the percentile interval is exact.

Proof. The exact interval is $\theta \in (\varphi^{-1}(\varphi(\hat\theta) - z_{\alpha/2}), \varphi^{-1}(\varphi(\hat\theta) + z_{\alpha/2}))$. In the bootstrap world $\varphi(\hat\theta^*) \sim N(\varphi(\hat\theta), 1)$, whose $\frac{\alpha}{2}$, $1-\frac{\alpha}{2}$ quantiles are $\varphi(\hat\theta) \pm z_{\alpha/2}$; φ^{-1} (increasing) maps them to $Q_{G^*}(\frac{\alpha}{2})$, $Q_{G^*}(1-\frac{\alpha}{2})$. φ is never computed. ■

It is transformation-respecting and stays in the parameter space; its weak point is assuming $\varphi(\hat\theta)$ median-unbiased with constant variance.

8.3.1 Bias-corrected and accelerated (BCa) intervals

Generalize the assumption to

$$\varphi(\hat\theta) - \varphi(\theta) \sim N(-b\,\sigma_\varphi,\ \sigma_\varphi^2), \qquad \sigma_\varphi = 1 + a\,\varphi(\theta), \tag{8.3}$$

with bias constant b and acceleration a (rate of change of the standard error with the parameter).

Proposition 8.2

Under (8.3) and its bootstrap analogue, the interval $(Q_{G^*}(p_l), Q_{G^*}(p_u))$ is exact, where with $z_l = \Phi^{-1}(\frac{\alpha}{2})$, $z_u = \Phi^{-1}(1-\frac{\alpha}{2})$,

$$p_l = \Phi\Big(b + \frac{b + z_l}{1 - a(b + z_l)}\Big), \qquad p_u = \Phi\Big(b + \frac{b + z_u}{1 - a(b + z_u)}\Big).$$

Proof [Supplement]. Write $\hat\varphi = \varphi(\hat\theta)$, $\phi = \varphi(\theta)$, $1 + a\phi > 0$. Then $Z = \frac{\hat\varphi - \phi}{1+a\phi} + b \sim N(0,1)$, and $\phi = g(Z) = \frac{\hat\varphi - (Z-b)}{1+a(Z-b)}$ with $g'(Z) = -\frac{1+a\hat\varphi}{(1+a(Z-b))^2} < 0$. So $\phi \in [g(-z_l), g(-z_u)] = [L, U]$ iff $Z \in [z_l, z_u]$ (probability $1-\alpha$), $L = \frac{\hat\varphi + b + z_l}{1 - a(b+z_l)}$, similarly U. In the bootstrap world $\varphi(\hat\theta^*) \sim N(\mu_*, \sigma_*^2)$, $\mu_* = \hat\varphi - b(1+a\hat\varphi)$, $\sigma_* = 1 + a\hat\varphi$. With $w = b + z_l$, $L - \hat\varphi = \frac{w(1+a\hat\varphi)}{1-aw}$, so $P^*(\varphi(\hat\theta^*) \le L) = \Phi(\frac{w}{1-aw} + b) = p_l$: $\varphi^{-1}(L) = Q_{G^*}(p_l)$; likewise U. ■

Estimating b and a. Under the model, $P^*(\hat\theta^* \le \hat\theta) = \Phi(b)$, so

$$\hat b = \Phi^{-1}\Big(\frac{\#\{\hat\theta^{*b} \le \hat\theta\}}{B}\Big).$$

For a, the jackknife ($\hat\theta_{(i)}$ without observation i, mean $\hat\theta_{(\cdot)}$):

$$\hat a = \frac{\sum_i (\hat\theta_{(\cdot)} - \hat\theta_{(i)})^3}{6\big[\sum_i (\hat\theta_{(\cdot)} - \hat\theta_{(i)})^2\big]^{3/2}},$$

a skewness of the influences. (The jackknife, Quenouille 1949, Tukey 1958, is a precursor of the bootstrap.) With $a = 0$ we get the **bias-corrected (BC)** interval, $p = \Phi(2\hat b + z)$.

8.4 Accuracy and recommendations

For a smooth statistic, $P(\theta \in C) = 1 - \alpha + O(n^{-k/2})$:

- first-order accurate ($k = 1$): normal theory, normal bootstrap, basic, percentile, BC;
- second-order accurate ($k = 2$): bootstrap-t, BCa.

Use bootstrap-t when a reliable SE formula exists, otherwise BCa, with $B \ge 2000$ (tail quantiles). The percentile interval is a simple default, often adequate for large samples. For multi-sample or regression resampling the jackknife $\hat a$ is less reliable.

8.5 Selected exercises

Motivation. Normalizing transformations always exist, basic vs. percentile under reparametrization, and two jackknife computations.

Exercise 8.2 (A normalizing transformation always exists)

Let $\hat{\theta}$ have a continuous, strictly increasing distribution function G (for the true θ), and put $\varphi(t) = \Phi^{-1}(G(t))$.
(a) Show that φ is increasing and $\varphi(\hat{\theta}) \sim N(0,1)$.
(b) If θ is the median of G, show that $\varphi(\hat{\theta}) - \varphi(\theta) \sim N(0,1)$.
(c) In general, with $b = \varphi(\theta)$, show that $\varphi(\hat{\theta}) - \varphi(\theta) \sim N(-b,1)$.

Solution.

(a) φ composes the increasing G and Φ^{-1}. By Exercise 4.6(a), $U = G(\hat{\theta}) \sim \text{Uniform}(0,1)$, and $P(\Phi^{-1}(U) \le z) = P(U \le \Phi(z)) = \Phi(z)$.

(b) $G(\theta) = \frac{1}{2}$ gives $\varphi(\theta) = \Phi^{-1}(\frac{1}{2}) = 0$; apply (a).

(c) $\varphi(\hat{\theta}) - \varphi(\theta) = Z - b$ with $Z = \varphi(\hat{\theta}) \sim N(0,1)$.

So the assumptions of Proposition 8.1 (case (b)) and of BC ((8.3) with $a = 0$, $\sigma_\varphi = 1$; case (c)) hold for some increasing φ whenever G is continuous. The real restriction: φ depends on θ and F, but the propositions need one φ valid also in the bootstrap world. ■

Exercise 8.7 (Basic versus percentile under reparametrization)

(a) Let g be a continuous, strictly increasing bijection between intervals containing the possible values of $\hat{\theta}$ and $\hat{\theta}^*$. Show that the percentile interval for $g(\theta)$, computed from the replicates $g(\hat{\theta}^{*b})$, equals g applied to the percentile interval for θ.
(b) Use the exact bootstrap distribution of Example 7.1 (data $\{1,4,9\}$, sample median) to compute the 60% basic and percentile intervals for the population median θ, and the basic interval for $\sqrt{\theta}$. Show that the basic interval does not have the property in (a).

Solution [Supplement].

(a) $P(g(T) \le g(x)) = P(T \le x)$, so $F_{g(T)}(g(x)) = F_T(x)$, and since g is an increasing bijection

$$Q_{g(T)}(p) = \min\{g(x) : F_T(x) \ge p\} = g(\min\{x : F_T(x) \ge p\}) = g(Q_T(p)).$$

Take $p = \frac{\alpha}{2}, 1 - \frac{\alpha}{2}$ and $T \sim G^*$.

(b)

Step 1. G^* puts mass $7/27, 13/27, 7/27$ on $1, 4, 9$; $\hat{\theta} = 4$. With $\alpha = 0.4$: $G^*(1) = 0.26 \ge 0.2$ gives $Q_{G^*}(0.2) = 1$; $G^*(4) = 0.74 < 0.8$ gives $Q_{G^*}(0.8) = 9$.

Step 2. Percentile interval for θ: $(1, 9)$. Basic interval (8.1): $(2 \cdot 4 - 9,\ 2 \cdot 4 - 1) = (-1, 7)$.

Step 3. For $\psi = \sqrt{\theta}$: $\hat{\psi} = 2$, replicates $1, 2, 3$ with the same probabilities. Percentile: $(1, 3)$, i.e. $(1, 9)$ for θ, as in (a). Basic: $(2 \cdot 2 - 3,\ 2 \cdot 2 - 1) = (1, 3)$, mapping back to $(1, 9) \ne (-1, 7)$.

So the basic interval depends on the scale; on the original scale it reflects the skew the wrong way and even goes negative. ■

Exercise 8.8 (The jackknife for two statistics)

Let $\hat\theta_{(i)}$ be the statistic computed without observation i and $\hat\theta_{(\cdot)} = \frac{1}{n}\sum_i \hat\theta_{(i)}$. The jackknife bias-corrected estimator and standard error are

$$\hat\theta_{\text{jack}} = n\hat\theta - (n-1)\hat\theta_{(\cdot)}, \qquad \widehat{\text{se}}_{\text{jack}} = \Big(\frac{n-1}{n}\sum_i (\hat\theta_{(i)} - \hat\theta_{(\cdot)})^2\Big)^{1/2}.$$

(a) For $\hat\theta = \hat\sigma^2 = \frac{1}{n}\sum_i (X_i - \bar X)^2$, show that $\hat\theta_{\text{jack}} = s^2$.
(b) For $\hat\theta = \bar X$, show that $\widehat{\text{se}}_{\text{jack}} = s/\sqrt{n}$.

Solution [Supplement]. Let $SS = \sum_j (X_j - \bar X)^2$ and $\bar X_{(i)} = (n\bar X - X_i)/(n-1)$, so $\bar X_{(i)} - \bar X = (\bar X - X_i)/(n-1)$.

(a)

Step 1. (Deleting one point.) Decomposing about $\bar X_{(i)}$,

$$\begin{aligned} SS_{(i)} &:= \sum_{j\neq i}(X_j - \bar X_{(i)})^2 = \sum_{j\neq i}(X_j - \bar X)^2 - (n-1)(\bar X_{(i)} - \bar X)^2 \\ &= SS - (X_i - \bar X)^2 - \frac{(X_i - \bar X)^2}{n-1} = SS - \frac{n}{n-1}(X_i - \bar X)^2. \end{aligned}$$

Step 2. $\hat\theta_{(i)} = SS_{(i)}/(n-1)$; summing over i,

$$\hat\theta_{(\cdot)} = \frac{1}{n(n-1)}\Big(nSS - \frac{n}{n-1}SS\Big) = \frac{(n-2)SS}{(n-1)^2}.$$

Step 3. Hence $\hat\theta_{\text{jack}} = SS - \frac{(n-2)SS}{n-1} = \frac{SS}{n-1} = s^2$: the bias of $\hat\sigma^2$ is removed exactly.

(b) $\hat\theta_{(i)} = \bar X_{(i)}$, $\hat\theta_{(\cdot)} = \bar X$, $\hat\theta_{(i)} - \hat\theta_{(\cdot)} = (\bar X - X_i)/(n-1)$, so

$$\widehat{\text{se}}^2_{\text{jack}} = \frac{n-1}{n}\cdot\frac{SS}{(n-1)^2} = \frac{SS}{n(n-1)} = \frac{s^2}{n}.$$

The ideal bootstrap gives the slightly smaller $\hat\sigma/\sqrt{n}$ (Exercise 7.5). ■

Chapter 9. Bootstrap Tests

The problem

A test needs the law of the statistic *under* H_0, but the bootstrap mimics the *actual* F. Two remedies:

1. **Resample under the null** from an estimate $\hat{F}_0$ satisfying H_0.
2. **Resample a pivot.** If $T_\theta = (\hat{\theta} - \theta)/\widehat{\text{se}}$ is approximately pivotal, approximate its law by $T^* = (\hat{\theta}^* - \hat{\theta})/\widehat{\text{se}}^*$ from $\hat{F}$ and compare with T_{θ_0}.

Either way $\hat{p} = \{1 + \#(T^{*b} \text{ at least as extreme as } T_{\text{obs}})\}/(B+1)$.

9.1 Parametric bootstrap tests

In a model $F(\cdot; \theta, \eta)$ with nuisance η, test $H_0 : \theta = \theta_0$ with a statistic $S(\mathbb{X}, \theta_0)$ that is large under H_1.

- *Null resampling:* restricted MLE $\hat{\eta}_0$ under $\theta = \theta_0$; draw from $F(\cdot; \theta_0, \hat{\eta}_0)$, recompute $S(\mathbb{X}^*, \theta_0)$.
- *Pivot resampling:* draw from $F(\cdot; \hat{\theta}, \hat{\eta})$ and recompute $S(\mathbb{X}^*, \hat{\theta})$.

Proposition 9.1

For $X_i \overset{\text{i.i.d.}}{\sim} N(\mu, \sigma^2)$ and $S = \sqrt{n}(\bar{X} - \mu_0)/s$, both schemes reproduce the exact null distribution t_{n-1}.

Proof. Null resampling: $N(\mu_0, \hat{\sigma}_0^2)$ samples give $\sqrt{n}(\bar{X}^* - \mu_0)/s^* \sim t_{n-1}$ for any $\hat{\sigma}_0$. Pivot resampling: $N(\hat{\mu}, \hat{\sigma}^2)$ samples give $\sqrt{n}(\bar{X}^* - \hat{\mu})/s^* \sim t_{n-1}$. ■

Exact pivots are rare. For Laplace(μ, b), $2\sum_i |X_i - \mu|/b \sim \chi^2_{2n}$ is a pivot only for known b; otherwise use approximate pivots (the t-statistic) or null resampling from Laplace$(\mu_0, \hat{b}_0)$, $\hat{b}_0 = \frac{1}{n}\sum |X_i - \mu_0|$. For two samples with a common scale, fit a common center (pooled median) and a pooled $\hat{b}_0$.

The schemes agree asymptotically; studentized statistics are more accurate than raw differences (Hall and Wilson). Use a non-pivotal statistic (e.g. $\hat{\theta}/\theta_0$ with θ-dependent spread) only with null resampling.

9.2 Nonparametric bootstrap tests

One sample, location. For $H_0 : \theta = \theta_0$ in $X_i = \theta + \epsilon_i$:

- null resampling from the recentered $\tilde{X}_i = X_i - \hat{\theta} + \theta_0$;
- pivot resampling from the raw data with $T^* = (\hat{\theta}^* - \hat{\theta})/\widehat{\text{se}}^*$.

For a mean with the t-statistic the two coincide (recentering shifts every bootstrap mean equally). Two-sided: $|T^*| \geq |T_{\text{obs}}|$.

Two samples, $H_0 : F = G$. Draw both samples from the pooled EDF: the with-replacement analogue of the permutation test, similar except in very small samples.

Two samples, equal locations, possibly different shapes. $F \neq G$ is allowed under $H_0 : \mu_1 = \mu_2$, so pooling and permutation are invalid. Use pivot resampling within each sample:

$$T^* = \frac{\bar{Y}^* - \bar{X}^* - (\bar{Y} - \bar{X})}{\sqrt{s_X^{*2}/m + s_Y^{*2}/n}}, \qquad T_{\text{obs}} = \frac{\bar{Y} - \bar{X}}{\sqrt{s_X^2/m + s_Y^2/n}}.$$

Equivalently, recenter each sample to the pooled mean and resample within samples. The *unstandardized* difference of recentered samples without subtracting $\bar{Y} - \bar{X}$ reproduces the sampling law of the observed difference, not a null law.

9.3 Duality with bootstrap intervals

Motivation. Every bootstrap interval yields a test by checking whether θ_0 falls inside.

Proposition 9.2

The two-sided level-α pivot test of $H_0 : \theta = \theta_0$ based on $\hat{\theta} - \theta$ (resp. on $(\hat{\theta} - \theta)/\widehat{\text{se}}$) accepts exactly when θ_0 lies in the basic (resp. bootstrap-t) interval.

Proof. Acceptance means $Q_{H^*}(\frac{\alpha}{2}) \leq \hat{\theta} - \theta_0 \leq Q_{H^*}(1 - \frac{\alpha}{2})$, i.e. $\hat{\theta} - Q_{H^*}(1 - \frac{\alpha}{2}) \leq \theta_0 \leq \hat{\theta} - Q_{H^*}(\frac{\alpha}{2})$: (8.1); likewise (8.2). ■

Percentile-type intervals have no simple test counterpart (φ is unknown).

9.4 Selected exercises

Motivation. An un-recentered bootstrap null, an exact Laplace pivot, and two-sample resampling with unequal variances.

Exercise 9.2 (A bootstrap test that never rejects)

Let $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} N(\theta, 1)$, treated as coming from an unknown distribution. For $H_0 : \theta = \theta_0$ against $H_1 : \theta > \theta_0$, take $S = \bar{X} - \theta_0$, let G^* be the bootstrap distribution of $S^* = \bar{X}^* - \theta_0$ (resampled from the raw data), and use the p-value $p = 1 - G^*(S_{\text{obs}})$. Decide whether: (a) G^* approximates the null distribution of S; (b) the test controls the type I error at $\alpha = 0.05$; (c) the power at $\theta_1 > \theta_0$ is reasonable, also as $\theta_1 \to \infty$.

Solution.

Step 1. (Where G^* is centered.) $S^* = (\bar{X}^* - \bar{X}) + S_{\text{obs}}$, and by Exercise 7.5(a) and the bootstrap CLT $\bar{X}^* - \bar{X} \approx N(0, \hat{\sigma}^2/n)$. So $G^* \approx N(S_{\text{obs}}, \hat{\sigma}^2/n)$ mimics $N(\theta - \theta_0, 1/n)$ around the observed value, not the null $N(0, 1/n)$: (a) fails unless $S_{\text{obs}} \approx 0$.

Step 2. (p does not depend on θ.) $p = P^*(\bar{X}^* > \bar{X})$ is shift invariant, a function of the residuals $x_i - \bar{x}$, whose law under $N(\theta, 1)$ is free of θ. So $P_{\theta_1}(p \leq \alpha) = P_{\theta_0}(p \leq \alpha)$ for every n and θ_1.

Step 3. (Size.) $p = P^*(\sqrt{n}(\bar{X}^* - \bar{X})/\hat{\sigma} > 0) \to \frac{1}{2}$ in probability, so $P(p \leq 0.05) \to 0$.

So (b) holds trivially (the test essentially never rejects) and (c) fails: power equals size for every θ_1. Remedy: recenter at θ_0, or compare the pivot $S^* - S_{\text{obs}} = \bar{X}^* - \bar{X}$ with S_{obs} (Section 9.2). ■

Exercise 9.5 (An exact Laplace pivot)

Let $X_1, \dots, X_n \overset{\text{i.i.d.}}{\sim} \text{Laplace}(\mu, b)$, with density $\frac{1}{2b}e^{-|x-\mu|/b}$, and b known.
(a) Show that $|X_i - \mu|/b \sim \text{Exp}(1)$ and $V(\mu) = 2\sum_i |X_i - \mu|/b \sim \chi^2_{2n}$.
(b) Show that $E|X - c| = d + be^{-d/b}$ with $d = |c - \mu|$, so $E|X - c|$ is uniquely minimized at $c = \mu$.
(c) Conclude that rejecting $H_0 : \mu = \mu_0$ when $V(\mu_0) > q_{1-\alpha}$, the $(1-\alpha)$-quantile of χ^2_{2n}, is an exact level-α test that is consistent against every $\mu \neq \mu_0$.

Solution [Supplement].

(a) With $Y = X - \mu$, $P(|Y|/b > t) = 2\int_{bt}^{\infty} \frac{1}{2b}e^{-y/b}\,dy = e^{-t}$; so $2|X_i - \mu|/b \sim \chi^2_2$ and the sum is χ^2_{2n} (Exercise 1.7(a)).

(b) Take $c = \mu + d$, $d \geq 0$. Use $|Y - d| = (Y - d) + 2(d - Y)^+$, $(d - Y)^+ = \int_{-\infty}^{d} I(Y \leq t)\,dt$, and $P(Y \leq t) = \frac{1}{2}e^{t/b}$ $(t < 0)$, $1 - \frac{1}{2}e^{-t/b}$ $(t \geq 0)$:

$$E(d - Y)^+ = \int_{-\infty}^{0} \tfrac{1}{2}e^{t/b}\,dt + \int_{0}^{d} (1 - \tfrac{1}{2}e^{-t/b})\,dt = d + \tfrac{b}{2}e^{-d/b}.$$

So $E|Y - d| = d + be^{-d/b}$, with derivative $1 - e^{-d/b} > 0$ for $d > 0$.

(c) Under H_0, $P(V(\mu_0) > q_{1-\alpha}) = \alpha$ by (a). Under $\mu \neq \mu_0$, $V(\mu_0)/(2n) \to E|X - \mu_0|/b > 1$ a.s. by the LLN and (b), while $q_{1-\alpha}/(2n) \to 1$; so $P(V(\mu_0) > q_{1-\alpha}) \to 1$. With b unknown $V(\mu_0)$ is not computable, hence Section 9.1. ■

Exercise 9.6 (Two samples with unequal variances)

Let $X_1, \dots, X_m$ have variance σ_X^2 and $Y_1, \dots, Y_n$ variance σ_Y^2, independent, with $H_0 : \mu_X = \mu_Y$.
(a) Recenter both samples at a common value c: $\tilde{X}_i = X_i - \bar{X} + c$, $\tilde{Y}_j = Y_j - \bar{Y} + c$. Show that, drawing the same resampling indices, the studentized statistic computed from $(\tilde{X}^*, \tilde{Y}^*)$ equals the pivot T^* of Section 9.2 computed from (X^*, Y^*).
(b) Show that resampling both samples from the pooled data gives, in large samples, $\text{Var}^*(\bar{Y}^* - \bar{X}^*) \approx \sigma_X^2/n + \sigma_Y^2/m$, whereas $\text{Var}(\bar{Y} - \bar{X}) = \sigma_X^2/m + \sigma_Y^2/n$.

Solution [Supplement].

(a) $\bar{\tilde{X}}^* = \bar{X}^* - \bar{X} + c$, $\bar{\tilde{Y}}^* = \bar{Y}^* - \bar{Y} + c$, so $\bar{\tilde{Y}}^* - \bar{\tilde{X}}^* = \bar{Y}^* - \bar{X}^* - (\bar{Y} - \bar{X})$, and sample variances are shift invariant: the statistics coincide replicate by replicate, for any c.

(b)

Step 1. With $N = m + n$, under H_0 the LLN gives the pooled empirical variance

$$\hat{\sigma}^2_{\text{pool}} \approx \frac{m\sigma_X^2 + n\sigma_Y^2}{N}.$$

Step 2. Both samples are drawn independently from the pooled EDF, so by Exercise 7.5(a)

$$\text{Var}^*(\bar{Y}^* - \bar{X}^*) = \hat{\sigma}^2_{\text{pool}}\Big(\frac{1}{m} + \frac{1}{n}\Big) \approx \frac{m\sigma_X^2 + n\sigma_Y^2}{N} \cdot \frac{N}{mn} = \frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}.$$

Step 3. The weights of σ_X^2, σ_Y^2 are swapped relative to $\text{Var}(\bar{Y} - \bar{X})$; they agree iff $m = n$ or $\sigma_X^2 = \sigma_Y^2$. Otherwise pooled resampling, like a permutation test, has the wrong null spread (anticonservative when the smaller sample has the larger variance), so resample within samples as in (a). ■

Chapter 10. Linear Regression

10.1 Simple linear regression

Motivation. Rank-based slopes such as the median pairwise slope are robust to outliers and give distribution-free intervals.

Model $y_i = \alpha + \beta x_i + \epsilon_i$ with uncorrelated errors of mean 0 and variance σ^2; inference is conditional on the x_i. Least squares gives $\hat\beta = S_{xy}/S_{xx}$, $\hat\alpha = \bar y - \hat\beta\bar x$ and $s^2 = \sum e_i^2/(n-2)$.

Proposition 10.1

$E\hat\beta = \beta$ and $\operatorname{Var}\hat\beta = \sigma^2/S_{xx}$. The residuals satisfy $\operatorname{Var}(e_i) = \sigma^2(1-h_i)$ with leverage $h_i = \frac{1}{n} + \frac{(x_i-\bar x)^2}{S_{xx}}$, so the adjusted residuals $r_i = e_i/\sqrt{1-h_i}$ have variance σ^2.

Proof. $\hat\beta = \sum w_i y_i$ with $w_i = (x_i - \bar x)/S_{xx}$, $\sum w_i = 0$, $\sum w_i x_i = 1$, $\sum w_i^2 = 1/S_{xx}$. With the hat matrix H, $\mathbf{e} = (I-H)\boldsymbol\epsilon$ and $\operatorname{Var}\mathbf{e} = \sigma^2(I-H)$, whose diagonal entries are $\sigma^2(1-h_i)$. ■

Under normal errors $(\hat\beta - \beta)/\widehat{\mathrm{se}}(\hat\beta) \sim t_{n-2}$, $\widehat{\mathrm{se}}(\hat\beta) = s/\sqrt{S_{xx}}$; otherwise resample.

Permutation test of $H_0 : \beta = 0$. Under H_0 the $y_i = \alpha + \epsilon_i$ are exchangeable: shuffle y against fixed x. $\hat\beta$, $\sum x_i y_i$ and $t = \hat\beta\sqrt{(n-2)S_{xx}}/\sqrt{S_{yy} - \hat\beta^2 S_{xx}}$ (increasing in $\hat\beta$) give the same test, the permutation test of correlation (Section 5.1).

Bootstrap intervals. Case bootstrap (random x, robust to model failure) or residual bootstrap $y_i^* = \hat\alpha + \hat\beta x_i + r_i^*$ (fixed design). Since $\widehat{\mathrm{se}}(\hat\beta)$ has a formula, the bootstrap-t interval

$$\beta \in \big(\hat\beta - \widehat{\mathrm{se}}(\hat\beta)Q_{t^*}(1-\tfrac{\alpha}{2}),\ \hat\beta - \widehat{\mathrm{se}}(\hat\beta)Q_{t^*}(\tfrac{\alpha}{2})\big), \qquad t^* = \frac{\hat\beta^* - \hat\beta}{\widehat{\mathrm{se}}(\hat\beta^*)},$$

is cheap and second-order accurate. Tests of $\beta = 0$:

- null resampling: draw x and y *independently* (case form), or $y^* = \bar y + r_0^*$, $r_{0,i} = (y_i - \bar y)/\sqrt{1-1/n}$ (residual form);
- pivot resampling: compare t^* with $t_{\mathrm{obs}} = \hat\beta/\widehat{\mathrm{se}}(\hat\beta)$ (inverting the bootstrap-t interval).

10.2 Multiple regression

$\mathbf{y} = X\boldsymbol\beta + \boldsymbol\epsilon$, X of size $n \times (p+1)$: $\hat{\boldsymbol\beta} = (X'X)^{-1}X'\mathbf{y}$, $E\hat{\boldsymbol\beta} = \boldsymbol\beta$, $\operatorname{Var}\hat{\boldsymbol\beta} = \sigma^2(X'X)^{-1}$, $s^2 = \mathbf{e}'\mathbf{e}/(n-p-1)$ (unbiased: $E\,\boldsymbol\epsilon'(I-H)\boldsymbol\epsilon = \sigma^2\mathrm{tr}(I-H)$). Each β_j is a *partial* effect, adjusted for the other regressors.

Comparing nested models. Split $X = [X_0, X_1]$ ($k+1$ and $q = p-k$ columns) and test $H_0 : \boldsymbol\beta_1 = 0$. With hat matrices H and H_0,

$$F = \frac{\mathbf{y}'(H - H_0)\mathbf{y}/q}{\mathbf{y}'(I-H)\mathbf{y}/(n-p-1)} = \frac{\hat{\boldsymbol\beta}_1'\{X_1'(I-H_0)X_1\}\hat{\boldsymbol\beta}_1}{q\,s^2}. \tag{10.1}$$

Proposition 10.2

Under H_0 with normal errors, $F \sim F_{q,n-p-1}$. More generally $F_{\boldsymbol{\beta}_1}$, defined by replacing $\hat{\boldsymbol{\beta}}_1$ with $\hat{\boldsymbol{\beta}}_1 - \boldsymbol{\beta}_1$ in (10.1), has this distribution for every $\boldsymbol{\beta}_1$, so it is a pivot.

Proof. $H - H_0$ is a rank-q projection orthogonal to $I - H$, both annihilating X_0. Under H_0 the forms reduce to $\boldsymbol{\epsilon}'(H - H_0)\boldsymbol{\epsilon}$ and $\boldsymbol{\epsilon}'(I - H)\boldsymbol{\epsilon}$, independent $\sigma^2\chi^2_q$ and $\sigma^2\chi^2_{n-p-1}$ (Cochran). The second form is Frisch–Waugh–Lovell: $\hat{\boldsymbol{\beta}}_1$ regresses $(I - H_0)\mathbf{y}$ on $\tilde{X}_1 = (I - H_0)X_1$, and $H - H_0$ projects onto $\mathrm{col}(\tilde{X}_1)$. For general $\boldsymbol{\beta}_1$ use $\mathbf{y} - X_1\boldsymbol{\beta}_1$. ■

For $q = 1$, $F = t_j^2$; a one-sided $\beta_j > 0$ needs t_j, to which the schemes below apply equally. Bootstrap null resampling: the case form (draw $\mathbf{x}$-rows and y's independently; when all slopes are tested) or reduced-model residuals with replacement. Pivot resampling: case or full-model residual bootstrap with $F_{\hat{\boldsymbol{\beta}}_1}$ or $t_j^* = (\hat{\beta}_j^* - \hat{\beta}_j)/\widehat{\mathrm{se}}(\hat{\beta}_j^*)$; p-value $\#\{F^{*b} \geq F_{\mathrm{obs}}\}/(B+1)$ (observed value counted).

Permutation tests for $H_0 : \boldsymbol{\beta}_1 = 0$. Under H_0 the errors are exchangeable but the y_i are not (means depend on X_0), so no permutation test is exact. Approximate schemes:

1. *Shuffle y:* effectively tests $\beta_1 = \cdots = \beta_p = 0$; can be anticonservative.
2. *Freedman–Lane:* shuffle reduced-model residuals e_0, $y^* = \hat{y}_0 + e_0^*$, recompute F; generally the most accurate.
3. *ter Braak:* shuffle full-model residuals, $y^* = \hat{y} + e^*$, recompute the pivot $F_{\hat{\boldsymbol{\beta}}_1}$.

All are asymptotically equivalent; sampling with replacement in 2 and 3 gives the null and pivot residual bootstrap tests. Bootstrap-t intervals for β_j use t_j^* with $\widehat{\mathrm{se}}^2(\hat{\beta}_j) = s^2(X'X)^{-1}_{jj}$.

10.3 Robust regression

Least squares (the normal MLE) loses efficiency under heavy tails and can be ruined by one outlier.

Definition (Breakdown point)

The smallest fraction of observations that, sent to infinity, can send the estimator to infinity: $1/n \to 0$ for the mean, α for the α-trimmed mean, $1/2$ for the median.

Trimmed and Winsorized means. The α-trimmed mean drops the lowest and highest α fractions and averages the rest; the α-Winsorized mean replaces them by the α, $1-\alpha$ sample quantiles and averages all n. Both break down at α. Hodges–Lehmann breaks down at $1 - 1/\sqrt{2} \approx 29\%$: a fraction ε of wild points drags $1 - (1-\varepsilon)^2$ of the Walsh averages, exceeding $\frac{1}{2}$ when $\varepsilon > 1 - 1/\sqrt{2}$.

Example 10.1

For $-2, -1, -1, 0, 0, 1, 1, 2, 3, 50$: mean 5.3; 10% trimmed mean 0.625 (drop -2 and 50); 10% Winsorized mean 0.7 (replace them by -1 and 3); median 0.5. Replacing 50 by 10^{100} changes only the mean.

Trimming and Winsorizing treat both tails alike; M-estimators down-weight by residual.

10.3.1 M-estimators

Generalize $\bar{x} = \arg\min_\mu \sum (x_i - \mu)^2$ to

$$\hat{\mu}^\Psi = \arg\min_\mu \sum_i \Psi\Big(\frac{x_i - \mu}{\hat{\tau}}\Big) \iff \sum_i \psi\Big(\frac{x_i - \mu}{\hat{\tau}}\Big) = 0, \qquad \psi = \Psi',$$

with robust scale $\hat{\tau} = 1.4826$ MAD (consistent for σ under normality, $\Phi^{-1}(3/4) = 0.6745$).

- $\psi(x) = x$: the mean. $\psi(x) = \mathrm{sgn}(x)$: the median.
- **Huber:** $\psi(x) = \max(-k, \min(k, x))$; $k \approx 1.35$ gives about 95% normal efficiency.
- **Tukey biweight:** $\psi(x) = x(1 - (x/c)^2)^2 I(|x| \le c)$, redescending: gross outliers get weight zero.

A bounded ψ with a 50% breakdown scale breaks down at 50%. ("M" = maximum-likelihood type: $\Psi(x) = x^2/2$, $\hat{\tau} = \sigma$ gives the normal MLE.) Solve numerically, e.g. by IRLS with weights $\psi(u_i)/u_i$.

Inference on a location with outliers. Outliers inflate both the mean and s, and so drive the t-test and t-interval. Robust alternatives:

- sign test and order-statistic interval (Chapter 2), valid for the median under any continuous F;
- signed-rank test and Walsh-average interval, if the uncontaminated law is symmetric;
- an M-estimator with bootstrap SE and bias (normal, percentile or BCa interval); its asymptotic SE needs the influence function, so the bootstrap is practical.

Lower breakdown tends to mean wider intervals under contamination. In regression,

$$\hat{\boldsymbol{\beta}}^\Psi = \arg\min_{\boldsymbol{\beta}} \sum \Psi((y_i - \mathbf{x}_i'\boldsymbol{\beta})/\hat{\tau}),$$

and the equations $\sum \psi(\cdot)\,\mathbf{x}_i = 0$ show Huber is robust to outlying y but not to high leverage (x_{ij} unbounded). **Yohai's MM-estimator**: redescending ψ (biweight, $c = 4.685$ for 95% efficiency) from a high-breakdown S-estimate of scale, giving 50% breakdown and high efficiency. **Least trimmed squares** minimizes the sum of the h smallest squared residuals ($h = \lfloor n/2 \rfloor + \lfloor (p+1)/2 \rfloor$ for maximal breakdown, $h = \lfloor 0.9n \rfloor$ for 10% trimming); **least median of squares** the median squared residual. Both are very robust, inefficient, and used as starting values.

10.3.2 The Theil–Sen estimator

For $y_i = \alpha + \beta x_i + \epsilon_i$, with pairwise slopes $\beta_{ij} = (y_j - y_i)/(x_j - x_i)$ $(x_i \ne x_j)$,

$$\hat{\beta}^{TS} = \mathrm{median}_{i<j}\, \beta_{ij}, \qquad \hat{\alpha}^{TS} = \mathrm{median}_i (y_i - \hat{\beta}^{TS} x_i).$$

Proposition 10.3 (Link to Kendall's tau)

Let $\tau_n(\beta_0)$ be Kendall's statistic between x_i and $y_i - \beta_0 x_i$. Then $\binom{n}{2}\tau_n(\beta_0) = \#\{\beta_{ij} > \beta_0\} - \#\{\beta_{ij} < \beta_0\}$. Hence the test of $H_0 : \beta = \beta_0$ by Kendall's tau, and the interval $(\beta_{(l)}, \beta_{(u)})$ with $l = \binom{n}{2}\frac{Q_K(\alpha/2)+1}{2}$, $u = \binom{n}{2}\frac{Q_K(1-\alpha/2)+1}{2}$ (Q_K the null quantile of τ_n), are exact. $\hat{\beta}^{TS}$ is the corresponding point estimate.

Proof. $\mathrm{sgn}[(x_i - x_j)\{(y_i - \beta_0 x_i) - (y_j - \beta_0 x_j)\}] = \mathrm{sgn}[(x_i - x_j)^2(\beta_{ij} - \beta_0)] = \mathrm{sgn}(\beta_{ij} - \beta_0)$. At the true β, $y_i - \beta x_i = \alpha + \epsilon_i$ is independent of x_i, so $\tau_n(\beta)$ has the null law K_n. The acceptance region

is a range of $\#\{\beta_{ij} > \beta_0\}$, a step function of β_0 dropping by one at each ordered slope, hence an interval of ordered slopes with the stated indices. ■

$\hat{\beta}^{TS}$ breaks down at $1-1/\sqrt{2} \approx 29\%$ and has about 96% ARE vs. least squares under normal errors.

Example 10.2

Points $(1,1),(2,2),(3,3.5),(4,4),(5,20)$, the last an outlier. Ordered slopes $0.5, 1, 1, 1, 1.25, 1.5, 4.75, 6, 8.25, 16$ give $\hat{\beta}^{TS} = 1.375$, $\hat{\alpha}^{TS} = -0.625$; least squares gives 4.0.

In multiple regression one version uses **backfitting**:

1. Start with single-variable Theil–Sen slopes $\hat{\beta}_j^{(0)}$ and set $\hat{\beta}_0^{(0)} = \text{median}_i(y_i - \sum_j \hat{\beta}_j^{(0)} x_{ij})$.
2. For $k = 1, 2, \ldots$ and each j, take the Theil–Sen slope of $y_i - \sum_{l \neq j} \hat{\beta}_l^{(k-1)} x_{il}$ on x_{ij}; update the intercept as a median.
3. Iterate until the coefficients stabilize.

Inference with robust regression estimators. M- and MM-estimates have approximate SEs from the final weighted LS step, reasonable for large n; bootstrap SEs are usually somewhat larger and more reliable. Use a case-resampling bootstrap-t interval $(\hat{\beta}_j - \widehat{\text{se}}\, Q_{t^*}(1-\frac{\alpha}{2}), \hat{\beta}_j - \widehat{\text{se}}\, Q_{t^*}(\frac{\alpha}{2}))$ or permutation tests of $t_j = \hat{\beta}_j / \widehat{\text{se}}(\hat{\beta}_j)$ (residual shuffling for one coefficient, Section 10.2). Repeated points in bootstrap samples can upset pairwise-slope estimators (ties in x); permutation tests avoid this.

10.4 Selected exercises

Motivation. An exact permutation test for any slope, the Frisch–Waugh–Lovell identity, and the Huber location estimator.

Exercise 10.1 (Permutation test of $\beta = \beta_0$)

In $y_i = \alpha + \beta x_i + \epsilon_i$ with fixed x_i and i.i.d. errors, describe a permutation test of $H_0 : \beta = \beta_0$ against $\beta \neq \beta_0$ for a given $\beta_0 \neq 0$, and show that it is exact.

Solution.

Step 1. (Why not shuffle y.) Under H_0, $Ey_i = \alpha + \beta_0 x_i$ varies with i, so the y_i are not exchangeable.

Step 2. (Transform.) $z_i = y_i - \beta_0 x_i = \alpha + (\beta - \beta_0) x_i + \epsilon_i$; under H_0, $z_i = \alpha + \epsilon_i$ are i.i.d. and unrelated to x_i.

Step 3. (Test.) Use $b(z) = S_{xz}/S_{xx} = \hat{\beta} - \beta_0$ (or its t-statistic) over the $n!$ (or M random) permutations of z: $p = \#\{\pi : |b(z_\pi)| \geq |b(z)|\}/n!$.

Step 4. (Exactness.) z is exchangeable under H_0 and permutations form a group, so Theorem 3.3 gives $P_{H_0}(p \leq \alpha) \leq \alpha$, with equality when $\alpha\, n!$ is an integer and there are no ties; for M random permutations use Proposition 3.4.

The non-rejected β_0 form an exact confidence set (Proposition 1.2). ■

Exercise 10.4 (Partial regression (Frisch–Waugh–Lovell))

Let $X = [X_0, X_1]$ have full column rank $p+1$, with q columns in X_1. Let $M_0 = I - H_0$, $\tilde{X}_1 = M_0 X_1$ (each column of X_1 replaced by its residuals on X_0) and $\mathbf{y}_{|0} = M_0\mathbf{y}$. Consider the partial regression $\mathbf{y}_{|0} = \tilde{X}_1\boldsymbol{\gamma} + \text{error}$. Show:
(a) its least squares estimator is $\hat{\boldsymbol{\gamma}} = \hat{\boldsymbol{\beta}}_1$, and its residual vector equals the full-model residual vector $\mathbf{e}$;
(b) the F-statistic for $\boldsymbol{\gamma} = 0$, computed with $n - p - 1$ residual degrees of freedom, equals (10.1).

Solution.

(a)

Step 1. The full fit is $\mathbf{y} = X_0\hat{\boldsymbol{\beta}}_0 + X_1\hat{\boldsymbol{\beta}}_1 + \mathbf{e}$ with $X_0'\mathbf{e} = X_1'\mathbf{e} = 0$.

Step 2. Apply M_0: $M_0X_0 = 0$ and $H_0\mathbf{e} = X_0(X_0'X_0)^{-1}X_0'\mathbf{e} = 0$, so $M_0\mathbf{e} = \mathbf{e}$ and

$$\mathbf{y}_{|0} = \tilde{X}_1\hat{\boldsymbol{\beta}}_1 + \mathbf{e}.$$

Step 3. $\tilde{X}_1'\mathbf{e} = X_1'M_0\mathbf{e} = X_1'\mathbf{e} = 0$, so $\boldsymbol{\gamma} = \hat{\boldsymbol{\beta}}_1$ solves the partial normal equations $\tilde{X}_1'(\mathbf{y}_{|0} - \tilde{X}_1\boldsymbol{\gamma}) = 0$.

Step 4. If $\tilde{X}_1\mathbf{c} = 0$ then $X_1\mathbf{c} = H_0X_1\mathbf{c} \in \mathrm{col}(X_0)$, so $\mathbf{c} = 0$ by full rank. Hence $\hat{\boldsymbol{\gamma}} = \hat{\boldsymbol{\beta}}_1$ uniquely, with residual $\mathbf{e}$.

(b) The partial RSS is $\mathbf{e}'\mathbf{e} = \mathbf{y}'(I-H)\mathbf{y}$; under $\boldsymbol{\gamma} = 0$ it is $\mathbf{y}'M_0\mathbf{y} = \mathbf{y}'(I - H_0)\mathbf{y}$. The difference is $\mathbf{y}'(H - H_0)\mathbf{y}$, so

$$F = \frac{\mathbf{y}'(H - H_0)\mathbf{y}/q}{\mathbf{y}'(I-H)\mathbf{y}/(n-p-1)},$$

which is (10.1). Naive software would use $n - q$ d.f.; the correct $n - p - 1$ is the dimension of $\mathrm{col}(X)^{\perp} \ni \mathbf{e}$. ■

Exercise 10.7 (The Huber location estimator)

Fix $\hat{\tau} > 0$ and $k > 0$, let $\psi(u) = \max(-k, \min(k, u))$, $u_i(\mu) = (x_i - \mu)/\hat{\tau}$ and $g(\mu) = \sum_i \psi(u_i(\mu))$.
(a) Show that the solutions of $g(\mu) = 0$ form a nonempty closed bounded interval. Show that it is a single point unless no observation lies within $k\hat{\tau}$ of the solutions.
(b) Show that every solution $\hat{\mu}$ satisfies $\hat{\mu} = \sum_i w_i x_i / \sum_i w_i$ with $w_i = \min\{1, k/|u_i(\hat{\mu})|\}$ (the fixed-point equation of iteratively reweighted least squares).
(c) Show that if $\max_i |x_i - \bar{x}| \le k\hat{\tau}$, then $\hat{\mu} = \bar{x}$.

Solution [Supplement].

(a)

Step 1. Each $\psi(u_i(\mu))$ is continuous and nonincreasing in μ, so is g; $g(\mu) = nk$ for $\mu \le \min_i x_i - k\hat{\tau}$ and $-nk$ for $\mu \ge \max_i x_i + k\hat{\tau}$.

Step 2. By the IVT g has a root; $g^{-1}(\{0\})$ is closed (continuity), bounded (Step 1), and an interval since $g(\mu_1) = g(\mu_2) = 0$ forces $g = 0$ on $[\mu_1, \mu_2]$ by monotonicity.

Step 3. If $g \equiv 0$ on $[\mu_1, \mu_2]$, $\mu_1 < \mu_2$, every nonincreasing term with constant sum is constant there; but a term with $|u_i(\mu)| < k$ at some $\mu \in (\mu_1, \mu_2)$ is strictly decreasing near μ. So a nondegenerate root interval needs every observation at distance $\ge k\hat{\tau}$ from (μ_1, μ_2) (equally many on each side); otherwise the root is unique.

(b) For $u_i \ne 0$, $\psi(u_i) = w_i u_i$ with $w_i = \min\{1, k/|u_i|\} \in (0, 1]$; for $u_i = 0$ take $w_i = 1$. Then $0 = \sum_i w_i(x_i - \hat{\mu})/\hat{\tau}$ and $\sum_i w_i > 0$ give $\hat{\mu} = \sum_i w_i x_i / \sum_i w_i$; points beyond $k\hat{\tau}$ get weight $\propto 1/|u_i|$.

(c) At $\mu = \bar{x}$ all $|u_i| \le k$, so $g(\bar{x}) = \sum_i (x_i - \bar{x})/\hat{\tau} = 0$. Suppose the root set were $[\mu_1, \mu_2] \ni \bar{x}$, $\mu_1 < \mu_2$. If all $x_i = \bar{x}$, $\mu \in (\mu_1, \mu_2)$ near $\bar{x}$ has $|u_i(\mu)| < k$; otherwise moving μ from $\bar{x}$ toward an observation on one side gives $|x_i - \mu| < |x_i - \bar{x}| \le k\hat{\tau}$. Both contradict Step 3, so $\hat{\mu} = \bar{x}$: Huber equals the mean unless some residual exceeds $k\hat{\tau}$. ■

Chapter 11. Smoothing and Density Estimation

11.1 From histograms to kernels

The EDF has no derivative, and a histogram depends on bin width and origin; averaging histograms over shifted origins gives the kernel estimator.

Definition (Kernel density estimator)

For a kernel K and bandwidth $h > 0$, $\hat{f}(x) = \dfrac{1}{nh}\sum_{i=1}^{n} K\Big(\dfrac{x - X_i}{h}\Big)$.

Standard kernels: $K \ge 0$, $\int K = 1$, $K(-u) = K(u)$, $\sigma_K^2 = \int u^2 K < \infty$, e.g. Gaussian ϕ, Epanechnikov $\frac{3}{4}(1-u^2)_+$, uniform $\frac{1}{2}I(|u| \le 1)$. Then $\hat{f}$ is a density. The choice of h matters far more than that of K.

Example 11.1

Data $\{0, 1, 3\}$, Gaussian kernel, $h = 1$: $\hat{f}(1) = \frac{1}{3}[\phi(1)+\phi(0)+\phi(2)] = \frac{1}{3}(0.242+0.399+0.054) = 0.232$.

11.2 Bias, variance and bandwidth

Motivation. Bias and variance determine the optimal bandwidth and the $n^{-4/5}$ rate.

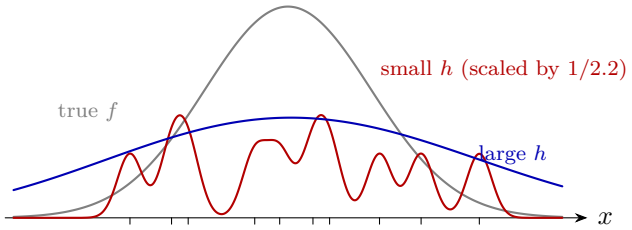


Figure. *Kernel estimates from ten points (ticks): small h is spiky (high variance; vertically shrunk), large h flattens the peak (bias).*

Theorem 11.1

Let f be twice continuously differentiable near x with $f(x) > 0$. As $n \to \infty$, $h \to 0$ and $nh \to \infty$,

$$\operatorname{bias}\hat{f}(x) = \frac{\sigma_K^2 f''(x)}{2}h^2 + o(h^2), \qquad \operatorname{Var}\hat{f}(x) = \frac{f(x)R(K)}{nh} + o\Big(\frac{1}{nh}\Big), \qquad R(K) = \int K^2.$$

Proof [Supplement]. $E\hat{f}(x) = \int K(u) f(x - hu)\,du$; expand

$$f(x - hu) = f(x) - huf'(x) + \tfrac{h^2u^2}{2}f''(x) + o(h^2u^2),$$

use $\int uK = 0$, $\int u^2 K = \sigma_K^2$ and dominated convergence. $\operatorname{Var}\hat{f}(x) = \frac{1}{n}\operatorname{Var}\big(\frac{1}{h}K(\frac{x-X}{h})\big)$, with $E\frac{1}{h^2}K^2(\frac{x-X}{h}) = \frac{1}{h}\int K^2(u) f(x-hu)\,du = \frac{f(x)R(K)}{h} + O(1)$ and squared mean $O(1)$. ∎

This is the **bias–variance trade-off**. Integrating,

$$\mathrm{MISE}(\hat f) = E\int (\hat f - f)^2 \approx \frac{\sigma_K^4 R(f'')}{4} h^4 + \frac{R(K)}{nh}, \qquad R(f'') = \int (f'')^2,$$

minimized at

$$h^* = \Big(\frac{R(K)}{\sigma_K^4 R(f'')\, n}\Big)^{1/5} \asymp n^{-1/5}, \qquad \mathrm{MISE}(h^*) \asymp n^{-4/5}. \tag{11.1}$$

(Set $\sigma_K^4 R(f'')h^3 - R(K)/(nh^2) = 0$.) The rate $n^{-4/5}$ is slower than the parametric n^{-1}.

Choosing h. h^* contains the unknown $R(f'')$.

- **Normal reference.** $f = N(\mu, \sigma^2)$ gives $R(f'') = \frac{3}{8\sqrt{\pi}\sigma^5}$, so with the Gaussian kernel $h = 1.06\,\hat\sigma n^{-1/5}$, $\hat\sigma = \min(s, \mathrm{IQR}/1.34)$.
- **Silverman's rule:** $h = 0.9\,\hat\sigma n^{-1/5}$, protecting against multimodality.
- **Plug-in (Sheather–Jones):** estimate $R(f'')$ by a pilot estimate of f'' in (11.1); generally recommended. Cross-validation is an alternative.

Proof of the normal value of $R(f'')$. For $\sigma = 1$, $\phi''(x) = (x^2-1)\phi(x)$ and $\phi^2 = \frac{1}{2\sqrt{\pi}}\times$ the $N(0, \frac12)$ density, so $R(\phi'') = \frac{1}{2\sqrt{\pi}} E(Z^2-1)^2 = \frac{1}{2\sqrt{\pi}}(\frac34 - 1 + 1) = \frac{3}{8\sqrt{\pi}}$, $Z \sim N(0, \frac12)$; scaling gives σ^{-5}. With $R(\phi) = \frac{1}{2\sqrt{\pi}}$, $\sigma_K = 1$, $h^* = (4/3)^{1/5}\sigma n^{-1/5} = 1.06\sigma n^{-1/5}$. ■

11.3 Multivariate density estimation

For $\mathbf{X}_i \in \mathbb{R}^p$, $\hat f(\mathbf{x}) = \frac1n \sum_i K_H(\mathbf{x} - \mathbf{X}_i)$, $K_H(\mathbf{x}) = |H|^{-1/2} K(H^{-1/2}\mathbf{x})$, with bandwidth matrix H (diagonal, or a full plug-in matrix). With a product kernel and equal widths h,

$$\mathrm{MSE}\,\hat f(\mathbf{x}) \approx \Big[\frac{\sigma_K^2 h^2}{2} \sum_j \partial_{jj} f(\mathbf{x})\Big]^2 + \frac{f(\mathbf{x}) R(K)^p}{nh^p},$$

so $h^* \asymp n^{-1/(p+4)}$, $\mathrm{MSE} \asymp n^{-4/(p+4)}$: the **curse of dimensionality** (for $p = 10$ a cube window of side 0.1 holds $10^{-10}n$ points on average). Practical for $p \le 2$, occasionally 3.

11.4 Confidence intervals and bands

At h^* bias and SE have the same order, so intervals target $f_h(x) = E\hat f(x)$.

- **Pointwise, plug-in:** by the CLT, $\sqrt{nh}(\hat f(x) - f_h(x)) \xrightarrow{d} N(0, f(x)R(K))$, so $f_h(x) \in \hat f(x) \pm z_{\alpha/2}\sqrt{\hat f(x) R(K)/(nh)}$.
- **Pointwise, bootstrap:** recompute $\hat f^*$ on a grid with the same h; use $\hat f \pm z_{\alpha/2}\widehat{\mathrm{se}}^*$, the basic $(2\hat f - Q^*_{1-\alpha/2}, 2\hat f - Q^*_{\alpha/2})$, or $\hat f \pm Q_{1-\alpha}(|\hat f^* - \hat f|)$.
- **Simultaneous band:** with q^* the $1-\alpha$ quantile of $\Delta^* = \max_{x\in\mathrm{grid}} |\hat f^*(x) - \hat f(x)|$, $\hat f(x) \pm q^*$ covers f_h on the grid with asymptotic probability $1-\alpha$; much wider.

Negative lower limits can be truncated at 0.

A bootstrap-t with $t^* = (\hat f^* - \hat f)/\sqrt{\hat f^* R(K)/(nh)}$ is unstable: where $\hat f^*(x) \approx 0$, t^* explodes; this also rules out studentized-maximum bands. Pointwise intervals on a grid are reported more often (some authors call them a "confidence band").

11.5 Selected exercises

Motivation. The $n^{-4/5}$ rate, why cross-validation targets MISE, and how little the kernel matters.

Exercise 11.5 (The $n^{-4/5}$ rate)

(a) Show that the asymptotic MISE at h^* in (11.1) equals $\frac{5}{4}\{\sigma_K^4 R(f'')\}^{1/5} R(K)^{4/5} n^{-4/5}$.
(b) For x with $f(x) > 0$ and f as in Theorem 11.1, show that MSE $\hat{f}(x) = O(n^{-4/5})$ for any $h = Cn^{-1/5}$.
(c) Compare with $\mathrm{MSE}(\bar{X}_n) = \sigma^2/n$.

Solution.

(a) Let $A(h) = ah^4 + c/(nh)$, $a = \sigma_K^4 R(f'')/4$, $c = R(K)$. At the minimizer $4ah^{*3} = c/(nh^{*2})$, i.e. $ah^{*4} = c/(4nh^*)$, so

$$A(h^*) = \frac{c}{4nh^*} + \frac{c}{nh^*} = \frac{5}{4} \cdot \frac{c}{nh^*}.$$

With $h^* = \{c/(4an)\}^{1/5}$, $A(h^*) = \frac{5}{4}(4a)^{1/5} c^{4/5} n^{-4/5}$, $4a = \sigma_K^4 R(f'')$.

(b) By Theorem 11.1,

$$\mathrm{MSE}\, \hat{f}(x) = \mathrm{bias}^2 + \mathrm{Var} = \Big(\frac{\sigma_K^2 f''(x)}{2}\Big)^2 h^4 + \frac{f(x) R(K)}{nh} + o\Big(h^4 + \frac{1}{nh}\Big).$$

With $h = Cn^{-1/5}$, $h^4 = C^4 n^{-4/5}$ and $1/(nh) = C^{-1} n^{-4/5}$.

(c) $\mathrm{MSE}\, \hat{f}(x) / \mathrm{MSE}(\bar{X}_n) \asymp n^{1/5} \to \infty$: halving the MSE needs $2^{5/4} \approx 2.38$ times the data, against 2 for a mean. ■

Exercise 11.7 (Least squares cross-validation)

Let $\hat{f}_{-i}(x) = \frac{1}{(n-1)h} \sum_{j \neq i} K\big(\frac{x - X_j}{h}\big)$ and

$$\mathrm{CV}(h) = \int \hat{f}^2 - \frac{2}{n} \sum_{i=1}^{n} \hat{f}_{-i}(X_i).$$

Assuming K is bounded and $\int f^2 < \infty$, show that $E\,\mathrm{CV}(h) = \mathrm{MISE}(h) - \int f^2$ for every h. Explain why the leave-one-out estimate is needed.

Solution [Supplement].

Step 1. $\mathrm{MISE}(h) = E\int \hat{f}^2 - 2E\int \hat{f} f + \int f^2$, and with $K_h(u) = K(u/h)/h$ and Fubini,

$$E\int \hat{f}(x) f(x)\, dx = \int E\hat{f}(x)\, f(x)\, dx = \iint K_h(x - t) f(t) f(x)\, dt\, dx.$$

Step 2. For $j \neq i$, $EK_h(X_i - X_j) = \iint K_h(x - t) f(x) f(t)\, dx\, dt$, so $E\hat{f}_{-i}(X_i)$ is the same integral and $E\big[\frac{2}{n}\sum_i \hat{f}_{-i}(X_i)\big] = 2E\int \hat{f} f$.

Step 3. So $E\,\mathrm{CV}(h) = \mathrm{MISE}(h) - \int f^2$, the last term free of h.

Using $\hat{f}(X_i) = \frac{K(0)}{nh} + \frac{n-1}{n}\hat{f}_{-i}(X_i)$ would subtract an extra $\frac{2K(0)}{nh} \to \infty$ as $h \to 0$, driving $h \to 0$: each point "predicted" by its own spike. ■

Exercise 11.8 (How much does the kernel matter?)

(a) Show that $\sigma_K R(K)$ is unchanged by rescaling $K_c(u) = K(u/c)/c$, and that the optimal AMISE is proportional to $\{\sigma_K R(K)\}^{4/5}$.
(b) Compute $\sigma_K R(K)$ for the Gaussian, Epanechnikov and uniform kernels, and the relative sample sizes needed for equal optimal AMISE.
(c) Derive the normal-reference bandwidth for the Epanechnikov kernel.

Solution [Supplement].

(a) With $u = cv$, $\sigma^2_{K_c} = c^2\sigma^2_K$ and $R(K_c) = R(K)/c$, so $\sigma_{K_c}R(K_c) = \sigma_K R(K)$: rescaling only changes the meaning of h. By Exercise 11.5(a) the optimal AMISE is

$$\frac{5}{4}R(f'')^{1/5}(\sigma_K^4)^{1/5}R(K)^{4/5}n^{-4/5} = \frac{5}{4}R(f'')^{1/5}\{\sigma_K R(K)\}^{4/5}n^{-4/5}.$$

(b) Gaussian: $\sigma_K = 1$, $R(K) = \frac{1}{2\sqrt{\pi}}$, product 0.2821. Epanechnikov: $\sigma^2_K = \frac{3}{4}\int_{-1}^{1}(u^2 - u^4)\,du = \frac{1}{5}$, $R(K) = \frac{9}{16}\int_{-1}^{1}(1-u^2)^2\,du = \frac{3}{5}$, product $\frac{3}{5\sqrt{5}} = 0.2683$. Uniform: $\sigma^2_K = \frac{1}{3}$, $R(K) = \frac{1}{2}$, product 0.2887. Equal AMISE needs $n \propto \sigma_K R(K)$: relative to Epanechnikov, Gaussian 1.05, uniform 1.08. The kernel costs at most a few percent.

(c) With $R(f'') = \frac{3}{8\sqrt{\pi}\sigma^5}$ (Section 11.2), (11.1) gives

$$h^* = \Big(\frac{3/5}{(1/25)\cdot 3/(8\sqrt{\pi}\sigma^5)\cdot n}\Big)^{1/5} = (40\sqrt{\pi})^{1/5}\sigma n^{-1/5} = 2.34\,\sigma n^{-1/5}.$$

It exceeds the Gaussian $1.06\,\sigma n^{-1/5}$ only because of scale ($\sigma_K = 0.447$ vs. 1), as (a) explains. ■

Chapter 12. Nonparametric Regression

Model

$Y_i = m(X_i) + \sigma(X_i)\epsilon_i$, $E\epsilon_i = 0$, $\operatorname{Var}\epsilon_i = 1$, with $m(x) = E(Y \mid X = x)$ smooth but otherwise unrestricted.

12.1 Local polynomial regression

Motivation. Fit a kernel-weighted low-degree polynomial around each point; local linear fits remove the boundary bias of local averages.

Definition (Local polynomial estimator)

With $K_h(u) = K(u/h)/h$, fit at each x

$$\hat{\boldsymbol{\beta}}_x = \arg\min_{\boldsymbol{\beta}} \sum_{i=1}^{n} \big(Y_i - \beta_0 - \beta_1(X_i - x) - \cdots - \beta_p(X_i - x)^p\big)^2 K_h(x - X_i),$$

and set $\hat{m}_p(x) = \hat{\beta}_{0,x}$. In matrix form, $\hat{\boldsymbol{\beta}}_x = (X_x' W_x X_x)^{-1} X_x' W_x \mathbf{y}$, with $W_x = \operatorname{diag}(K_h(x - X_i))$ and X_x having rows $(1, X_i - x, \ldots, (X_i - x)^p)$.

Each local polynomial estimator is linear: $\hat{m}_p(x) = \sum_i w_i^p(x) Y_i$.

Proposition 12.1

With $S_q = \sum_j K_h(x - X_j)(X_j - x)^q$:

(i) $p = 0$ (**Nadaraya–Watson**): $w_i^0(x) = K_h(x - X_i)/\sum_j K_h(x - X_j)$;

(ii) $p = 1$ (**local linear**): $w_i^1(x) = \dfrac{\{S_2 - (X_i - x)S_1\}K_h(x - X_i)}{S_0 S_2 - S_1^2}$.

In both cases $\sum_i w_i(x) = 1$. For $p = 1$ also $\sum_i w_i^1(x)(X_i - x) = 0$: the fit reproduces straight lines exactly.

Proof [Supplement]. (i) Minimize $\sum (Y_i - \beta_0)^2 K_h(x - X_i)$.

(ii) $X_x' W_x X_x = \begin{pmatrix} S_0 & S_1 \\ S_1 & S_2 \end{pmatrix}$, whose inverse has first row $(S_2, -S_1)/(S_0 S_2 - S_1^2)$; multiply by $X_x' W_x$. Then $\sum_i (S_2 - (X_i - x)S_1)K_h = S_2 S_0 - S_1^2$ and $\sum_i (S_2 - (X_i - x)S_1)(X_i - x)K_h = S_2 S_1 - S_1 S_2 = 0$. ■

As $h \to \infty$, Nadaraya–Watson tends to $\bar{Y}$ and local linear to the LS line.

Theorem 12.2

Let m be twice continuously differentiable, let the design density f be smooth with $f(x) > 0$, and use the Gaussian kernel. As $n \to \infty$, $h \to 0$ and $nh \to \infty$, conditionally on the X's and for $p = 0, 1$,

$$\operatorname{bias}\hat{m}_p(x) = B_p(x)h^2 + o_P(h^2), \qquad \operatorname{Var}\hat{m}_p(x) = \frac{\sigma^2(x)}{2\sqrt{\pi} f(x)} \cdot \frac{1}{nh} + o_P\Big(\frac{1}{nh}\Big),$$

with $B_0 = \dfrac{m''}{2} + \dfrac{m'f'}{f}$ and $B_1 = \dfrac{m''}{2}$.

Sketch. Since $\sum_i w_i = 1$, bias $= \sum_i w_i(x)\{m(X_i) - m(x)\}$ and Var $= \sum_i w_i^2 \sigma^2(X_i)$. For $p = 0$ the numerator $\frac{1}{n}\sum K_h(x - X_i)\{m(X_i) - m(x)\}$ has mean $\int K(u)\{m(x - hu) - m(x)\}f(x - hu)\,du = h^2(\frac{m''f}{2} + m'f') + o(h^2)$ and the denominator tends to $f(x)$, giving B_0. Variance: $\frac{1}{n^2}\sum K_h^2\sigma^2 \approx \frac{\sigma^2 f}{nh}R(K)$, $R(\phi) = \frac{1}{2\sqrt{\pi}}$, divided by f^2. For $p = 1$, $\sum w_i^1(X_i - x) = 0$ removes the term $m'(x)(X_i - x)$ exactly, including the design asymmetry behind $m'f'/f$; the variance is unchanged. ■

So local linear has the Nadaraya–Watson variance without the **design bias** $m'f'/f$, and behaves better at the boundary: the default. Bias decreases with degree, $|\text{bias}\,\hat{m}_0| \gtrsim |\text{bias}\,\hat{m}_1| \gtrsim |\text{bias}\,\hat{m}_2| \gtrsim \ldots$, while

$$\text{Var}\,\hat{m}_0 \approx \text{Var}\,\hat{m}_1 < \text{Var}\,\hat{m}_2 \approx \text{Var}\,\hat{m}_3 < \cdots .$$

so odd degrees ($p = 1$, sometimes 3) are preferred. The bias $\frac{m''}{2}h^2$ is positive in valleys and negative at peaks: features are flattened.

12.2 Choosing the bandwidth

Motivation. Cross-validation estimates prediction error, from a single fit for linear smoothers.

Integrating bias and variance against the design density,

$$\text{MISE} \approx \frac{h^4}{4}\int (m'')^2 f + \frac{\int \sigma^2}{2\sqrt{\pi}\,nh}, \qquad h^* = \Big(\frac{\int \sigma^2}{2\sqrt{\pi}\int (m'')^2 f \cdot n}\Big)^{1/5}.$$

- **Direct plug-in** (Ruppert–Sheather–Wand): estimate $\int (m'')^2 f$ and $\int \sigma^2$ by blockwise quartic fits.
- **Leave-one-out cross-validation:** minimize $\text{CV}(h) = \frac{1}{n}\sum_i \{Y_i - \hat{m}^{(-i)}(X_i)\}^2$.

Proposition 12.3

For a linear smoother whose leave-one-out fit equals the full fit at X_i after deleting point i (true for local polynomial fits), $\text{CV}(h) = \dfrac{1}{n}\sum_i \Big(\dfrac{Y_i - \hat{m}(X_i)}{1 - w_i(X_i)}\Big)^2$.

Proof [Supplement]. Let $\tilde{Y}_i = \hat{m}^{(-i)}(X_i)$. Replacing Y_i by $\tilde{Y}_i$ leaves the fit at X_i unchanged (the added point lies on the leave-one-out fit), so $\tilde{Y}_i = \sum_{j \neq i} w_j(X_i)Y_j + w_i(X_i)\tilde{Y}_i$, i.e. $\tilde{Y}_i = \frac{\hat{m}(X_i) - w_i(X_i)Y_i}{1 - w_i(X_i)}$ and $Y_i - \tilde{Y}_i = \frac{Y_i - \hat{m}(X_i)}{1 - w_i(X_i)}$. ■

So CV needs one fit per h.

12.3 Confidence intervals

Intervals target $m_h(x) = E\hat{m}_1(x)$.

- **Theory-based:** $\widehat{\text{se}}(\hat{m}_1(x)) = \big(\sum_i w_i^1(x)^2\hat{\sigma}^2(X_i)\big)^{1/2}$, $m_h(x) \in \hat{m}_1(x) \pm z_{\alpha/2}\widehat{\text{se}}$. Homoscedastic, with smoother matrix W ($\hat{\mathbf{y}} = W\mathbf{y}$):

$$\hat{\sigma}^2 = \frac{\|(I - W)\mathbf{y}\|^2}{\text{tr}\{(I - W)'(I - W)\}}.$$

  Heteroscedastic: smooth $\log r_i^2$ on X_i and exponentiate.

- **Bootstrap:** resample pairs, refit with the *same* h on a grid; use $\hat{m}_1(x) \pm z_{\alpha/2}\widehat{\mathrm{se}}^*(x)$ or $(Q^*_{\alpha/2}(x), Q^*_{1-\alpha/2}(x))$. Residual bootstraps need care (residuals inherit smoothing bias).

Simultaneous bands (Section 11.4) are rarely used.

12.4 Selected exercises

Motivation. Infinite-bandwidth limits, exact unbiasedness of local linear fits for lines, and when $\hat{\sigma}^2$ is unbiased.

Exercise 12.1, 12.4 (Infinite bandwidth)

Let K be continuous at 0 with $K(0) > 0$ (e.g. Gaussian), and suppose the X_i are not all equal. Show that as $h \to \infty$, for every x:
(a) the Nadaraya–Watson estimate tends to $\bar{Y}$;
(b) the local linear estimate tends to the least squares line $\bar{Y} + \hat{\beta}(x - \bar{X})$, $\hat{\beta} = S_{XY}/S_{XX}$.

Solution.

Step 1. (Scale invariance.) In Proposition 12.1, w_i^0 and w_i^1 are ratios of homogeneous expressions (degrees 1 and 2) in the $K_h(x - X_j)$, so replace these by $\kappa_j(h) = K((x - X_j)/h) \to K(0) > 0$.

(a)

Step 2. $w_i^0 = \kappa_i / \sum_j \kappa_j \to K(0)/(nK(0)) = 1/n$, so $\hat{m}_0(x) \to \bar{Y}$.

(b)

Step 3. With $d_j = X_j - x$, $S_q \to K(0) \sum_j d_j^q$ and the denominator tends to

$$K(0)^2 \Big\{ n \sum_j d_j^2 - \Big(\sum_j d_j \Big)^2 \Big\} = K(0)^2 \, n S_{XX} > 0.$$

So $w_i^1 \to (\sum_j d_j^2 - d_i \sum_j d_j)/(nS_{XX})$.

Step 4. With $\delta = \bar{X} - x$: $\sum_j d_j = n\delta$, $\sum_j d_j^2 = S_{XX} + n\delta^2$, $\sum_i d_i Y_i = S_{XY} + n\delta\bar{Y}$, so

$$\sum_i w_i^1 Y_i \to \frac{(S_{XX} + n\delta^2)\, n\bar{Y} - n\delta(S_{XY} + n\delta\bar{Y})}{nS_{XX}} = \bar{Y} - \delta \frac{S_{XY}}{S_{XX}} = \bar{Y} + \hat{\beta}(x - \bar{X}).$$

■

Exercise 12.5 (No bias for straight lines)

Suppose $m(x) = \alpha + \beta x$. Show that, conditionally on the X's, the local linear estimator is exactly unbiased for every h and x, while the Nadaraya–Watson estimator has bias $\beta \sum_i w_i^0(x)(X_i - x)$, which is nonzero in general (e.g. at $x = \min_i X_i$). Relate this to B_0 and B_1 in Theorem 12.2.

Solution.

Step 1. The weights depend only on the X's, so with $\sum_i w_i^p = 1$

$$E(\hat{m}_p(x) \mid X) - m(x) = \sum_i w_i^p(x) m(X_i) - m(x) = \beta \sum_i w_i^p(x)(X_i - x),$$

Step 2. For $p = 1$, $\sum_i w_i^1(x)(X_i - x) = 0$ by Proposition 12.1, so the bias is exactly 0.

Step 3. For $p = 0$ the bias is $\beta \sum_i K_h(x - X_i)(X_i - x) / \sum_i K_h(x - X_i)$. At $x = \min_i X_i$ all $X_i - x \geq 0$, some > 0, so the bias has the sign of β: the fit is pulled toward the interior.

Consistent with Theorem 12.2: $B_1 = m''/2 = 0$ for a line, while $B_0 = \beta f'/f$ is the design bias. ■

Exercise 12.6 (When is $\hat{\sigma}^2$ unbiased?)

Let $\hat{\mathbf{y}} = W\mathbf{y}$ be a linear smoother with $E(\mathbf{y} \mid X) = \mathbf{m} = (m(X_i))_i$ and $\mathrm{Var}(\mathbf{y} \mid X) = \sigma^2 I$. Show that

$$E(\|(I - W)\mathbf{y}\|^2 \mid X) = \|(I - W)\mathbf{m}\|^2 + \sigma^2 \mathrm{tr}\{(I - W)'(I - W)\}.$$

Deduce that the estimator $\hat{\sigma}^2$ of Section 12.3 is unbiased when m is linear and W is the local linear smoother, and biased upward otherwise.

Solution [Supplement].

Step 1. With $\mathbf{y} = \mathbf{m} + \boldsymbol{\epsilon}$ and $A = (I - W)'(I - W)$,

$$\|(I - W)\mathbf{y}\|^2 = \|(I - W)\mathbf{m}\|^2 + 2\mathbf{m}'A\boldsymbol{\epsilon} + \boldsymbol{\epsilon}'A\boldsymbol{\epsilon}.$$

Step 2. The middle term has mean 0, and $E\boldsymbol{\epsilon}'A\boldsymbol{\epsilon} = E\mathrm{tr}(A\boldsymbol{\epsilon}\boldsymbol{\epsilon}') = \sigma^2 \mathrm{tr}A$.

Step 3. So $E\hat{\sigma}^2 = \sigma^2 + \|(I - W)\mathbf{m}\|^2 / \mathrm{tr}A \geq \sigma^2$. Row i of W holds the weights of the fit at X_i; for linear m and local linear W, Exercise 12.5 gives $(I - W)\mathbf{m} = 0$, so $\hat{\sigma}^2$ is unbiased. Otherwise smoothing bias inflates it. ■